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Thermal–electrical modeling and experimental study of rainwater harvesting and active cooling for rooftop photovoltaic modules

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ABSTRACT

As rooftop photovoltaic systems typically operate at elevated temperatures, effective thermal management to improve their performance has become an important focus. This study presents an integrated analysis of rooftop photovoltaics, combining the development of a dynamic thermal–electrical coupled model incorporating active water cooling with rainwater harvesting experiments. The model captures transient heat transfer processes in multilayer photovoltaic modules and integrates the time-varying characteristics of water flow rate, temperature, and tank volume to simulate the complete water-cooling cycle. Model validation demonstrates high simulation accuracy for both power and module temperature, with NRMSE below 5% under cooled and uncooled scenarios. Parametric analysis shows that the maximum net power gain ratio from water cooling relatively increases by approximately 52% with increasing solar irradiance, while higher wind speed reduces it by up to 19.5%. Seasonal evaluation under clear-sky summer conditions indicates peak hourly net power gains of up to 7.0% of the rated module capacity and daily energy gains exceeding 100 Wh per module, with evaporative water consumption ranging from 3.33 to 1.64 L/day. Rainwater harvesting experiments further show that modules achieve collection efficiencies exceeding 85% under favorable wind directions, with summer rainwater harvesting volumes approximately two to three times greater than the cooling water consumption, supporting the practical feasibility of integrating rainwater resources with photovoltaic cooling strategies.

1. Introduction

Amid growing global demand for sustainable energy, solar photovoltaics has become a key power generation technology due to its clean, efficient, and sustainable nature [1]. To date, silicon-based photovoltaic cells continue to be the most widely utilized material among various types of photovoltaic cells [2]. However, the conversion efficiency of commercial silicon-based photovoltaic modules remains limited, typically around 20%, and can decline further under practical environmental conditions [3].

A key factor limiting the performance of photovoltaic modules is the rise in operating temperature. Since only a small fraction of the incident solar energy is converted into electricity, the remainder accumulates as thermal energy within the module, leading to temperature-induced efficiency degradation [4,5]. For silicon-based modules, the conversion efficiency typically decreases by 0.30–0.50% for every 1 °C rise above 25 °C [6–8]. To mitigate overheating losses, thermal management strategies for photovoltaic modules are generally classified into passive and active cooling approaches [9]. Passive cooling methods, such as the integration of phase change materials [10] or the installation of heat sinks [11,12], remove heat without additional energy consumption. However, their effectiveness is often constrained by ambient environmental conditions. In contrast, active cooling methods require external power to drive cooling fluids (air or water), thereby enabling superior temperature control and system stability [13,14]. Among these, water-based systems generally outperform air-based ones due to higher thermal conductivity and more favorable energy efficiency [15].

In recent years, extensive studies have been carried out to investigate and validate the performance enhancement of photovoltaic modules through active water-based cooling techniques. Nizer et al. [16] experimentally demonstrated that active water cooling can enhance power output by up to 16.3% and reduce panel temperature significantly under peak irradiance. Subsequently, Hadipour et al. [17] optimized the water

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Nomenclature

Symbols

A	area (m^2)
A_E	effective catchment area (m^2)
C	thermal capacitance (J/K)
c_p	specific heat (J/(kg K))
D	pipe diameter (m)
D_v	diffusion coefficient of water vapor in air (m^2/s)
H	photovoltaic module length (m)
H_R	rainfall (m)
h	heat transfer coefficient (W/m^2K)
h_m	mass transfer coefficient (m/s)
I	current (A)
G	incident solar radiation on the module plane (W/m^2)
Gr	Grashof number (dimensionless)
g	gravity acceleration (m/s^2)
J_R	average rainfall intensity (mm/h)
k	thermal conductivity (W/(m·K))
L	characteristic length (m)
L_v	latent heat of vaporization (J/kg)
m	mass (kg)
$\dot{m}$	mass flow rate (kg/s)
n	number of data points
P	power (W)
p_a	saturated vapor pressure at the ambient temperature (Pa)
p_w	saturated vapor pressure at the water-film surface (Pa)
Q_F	water flow rate (m^3/s)
Q	heat transfer (W)
q	electronic charge (1.60218×10^{-19}C)
R	electrical/thermal resistance (Ω, K/W)
Ra	Rayleigh number (dimensionless)
Re	Reynolds number (dimensionless)
RH	Relative humidity (%)
R_v	gas constant for water vapor (461.5 J/(kg·K))
Sc	Schmidt number (dimensionless)
Sh	Sherwood number (dimensionless)
T	temperature (℃, K(modeling))
U	voltage (V)
V	volume (m^3)
v	fluid velocity (m/s)
W	photovoltaic module width (m)
Y_c	simulated value
Y_m	measured value
Z	head (m)
α	absorptivity (dimensionless)
α_d	thermal diffusivity (m^2/s)
β	coefficient of isobaric volume expansion (1/K)
δ	thickness (m)
ε	emissivity (dimensionless)
μ	dynamic viscosity (Pa·s)
η	efficiency (%)
υ	kinematic viscosity (m^2/s)
θ	module tilt angle (∘)
ϕ	absorbed solar radiation (W/m^2)
ρ	density (kg/m^3)
$\rho_{v,a}$	saturated vapor densities at the ambient temperature (kg/m^3)
$\rho_{v,w}$	saturated vapor densities at the water-film surface (kg/m^3)
σ	Stefan-Boltzmann constant (5.67×10^{-8} $W/(m^2 \cdot K^4)$)
τ	transmissivity (dimensionless)
ξ	compensation coefficient (dimensionless)
ξ_I	current temperature coefficient (%/∘C)
ξ_U	voltage temperature coefficient (%/∘C)

Subscripts

a	ambient
BS	back sheet layer
C	solar cells layer
conv	convection
evap	evaporation
EVA	EVA (Ethylene Vinyl Acetate) layer
forced	forced convection
free	natural convection
g	glass layer
gr	roof and surrounding walls
m	maximum power point
NCC	no cooling condition
oc	open circuit
pump	water pump
rad	radiation
rain	rainwater harvesting
rat	net gain
ref	reference at standard test condition
sc	short circuit
sh	shunt
sky	sky
tk	water tank body
WCC	water cooling condition
w	water
wf	water film

spray cooling experimental strategy, achieving a 27.7% increase in the maximum power output of photovoltaic modules and enhancing their economic performance. Beyond experiments, numerical and data-driven modeling efforts have deepened the understanding of active water-based cooling [14]. Bevilacqua et al. [18] developed and validated a thermal model for rooftop photovoltaic systems, showing that active water-based cooling can lower cell temperature by 28.2% and boost power output by 7.8%.

Similarly, Da Silva et al. [19] developed a nonlinear thermal model to evaluate water-film cooled photovoltaic modules, reporting up to 35% temperature drops and 5–12% power gains, effectively capturing the cooling impact on performance. Azimi et al. [20] proposed a two-step optimization strategy for water-film cooling of photovoltaic modules, using a steady-state thermal model to guide the adjustment of flow rate and activation timing by ambient conditions, boosting efficiency over 14.8%. Additionally, Lopez-Pascual et al. [3] proposed and validated a geothermal-based closed-loop water cooling system, supported by a thermoelectrical model assuming thermal steady conditions, which demonstrated a peak monthly efficiency gain of 10.7% and an annual average power increase of 4.9%. While these studies advanced the understanding of active water cooling, most models focus on steady-state conditions, with limited consideration of dynamic changes in coolant properties. In particular, the potential influence of cooling water temperature rise and gradual consumption during operation on thermal-electrical interactions has yet to be fully addressed, which may affect the fidelity of performance predictions under realistic cooling scenarios.

The critical issue of water consumption in active water-based cooling of photovoltaic modules has prompted research efforts to combine power generation gains with efficient water-use strategies. Shahverdian et al. [21] quantified the power gain and determined the flow rate that minimizes evaporation loss based on a thermal model of photovoltaic active cooling. Subsequent studies applied dynamic multi-objective

optimization to maximize power output while reducing water consumption, with a 41.98% annual reduction in water consumption compared to constant-flow cooling [22]. Yang et al. [23] developed a water-based photovoltaic cooling system with a U-shaped borehole heat exchanger, incorporating a performance model to assess system behavior, where the near-closed circulation loop notably reduced water loss during the cooling process. Similarly, Zhang et al. [24] simulated heat and mass transfer in a semi-enclosed spray cooling system, demonstrating significant temperature reduction and suppressed water evaporation. Ismael et al. [25] designed a hybrid fin-assisted spray cooling system that activates water cooling on demand, effectively reducing water loss and enhancing energy efficiency and sustainability.

While these studies enhance water-use efficiency in photovoltaic active cooling, they primarily depend on external water sources without addressing water resource availability. This reliance can be particularly limiting in areas where external water supply is scarce or difficult to access [26]. Photovoltaic modules themselves, with their inclined and smooth surfaces, provide favorable structural conditions for rainwater harvesting [27,28]. García-Chica et al. [29] developed a PV-based rainwater harvesting system primarily aimed at supporting sustainable irrigation, showing significant potential for water resource management. Chekired et al. [30] integrated rainwater harvesting into an agrivoltaic system design, demonstrating that photovoltaic modules can serve as rainwater catchment surfaces while supporting electricity generation and agricultural water use. Arshad et al. [31] assessed the rainwater harvesting potential of photovoltaic installations in the Sahel region at approximately 800 mm/year, highlighting their benefits for improving water supply in arid regions. Alazzam et al. [32] experimentally investigated rainwater harvesting from solar panels in Jordan and indicated that photovoltaic surfaces can act as effective catchment areas, providing supplementary water resources for groundwater recharge. However, these studies mainly focus on water supply and resource management applications, while the potential use of harvested rainwater as a cooling medium for photovoltaic modules has received little attention. In particular, the relationship between the rainwater harvesting characteristics of photovoltaic modules and their potential cooling water demand has not been systematically evaluated, highlighting the potential for research into using in-situ rainwater harvesting to assist photovoltaic cooling.

In light of the identified research limitations, this study conducts an integrated analysis of rooftop photovoltaics, combining the development of a dynamic thermal–electrical coupled model incorporating active water cooling with rainwater harvesting experiments. The model is developed to comprehensively capture the transient response of multilayer heat transfer and power output during the active water-cooling cycle. Through model-based analysis, the influence of environmental conditions on the cooling performance of photovoltaic modules is systematically investigated, and the seasonal cooling gains and evaporation demand are evaluated. Rainwater harvesting experiments further quantify the rainwater collection capacity of rooftop photovoltaic modules and analyze their rainwater harvesting characteristics, exploring the role of self-collected water in the cooling management of photovoltaics. Together, the model and experimental results provide valuable insights into the performance of active water-cooled photovoltaic systems and the feasibility of using self-collected rainwater to support sustainable cooling.

2. Dynamic thermal–electrical model incorporating active water-cooling effects for photovoltaic modules

Fig. 1 presents a schematic of the photovoltaic module integrated with the active cooling system. The system includes photovoltaic modules mounted on a support frame with an adjustable inclination, a PT100 temperature sensor, an inverter, a data collector, a water collection trough, a water tank, a water pump, water return and outlet pipes, and environmental factors monitored. During operation, cooling water is pumped from the tank to form a thin water film on the module surface. This water film absorbs heat from the module and then returns to the tank.

Building on this system configuration, a dynamic coupled

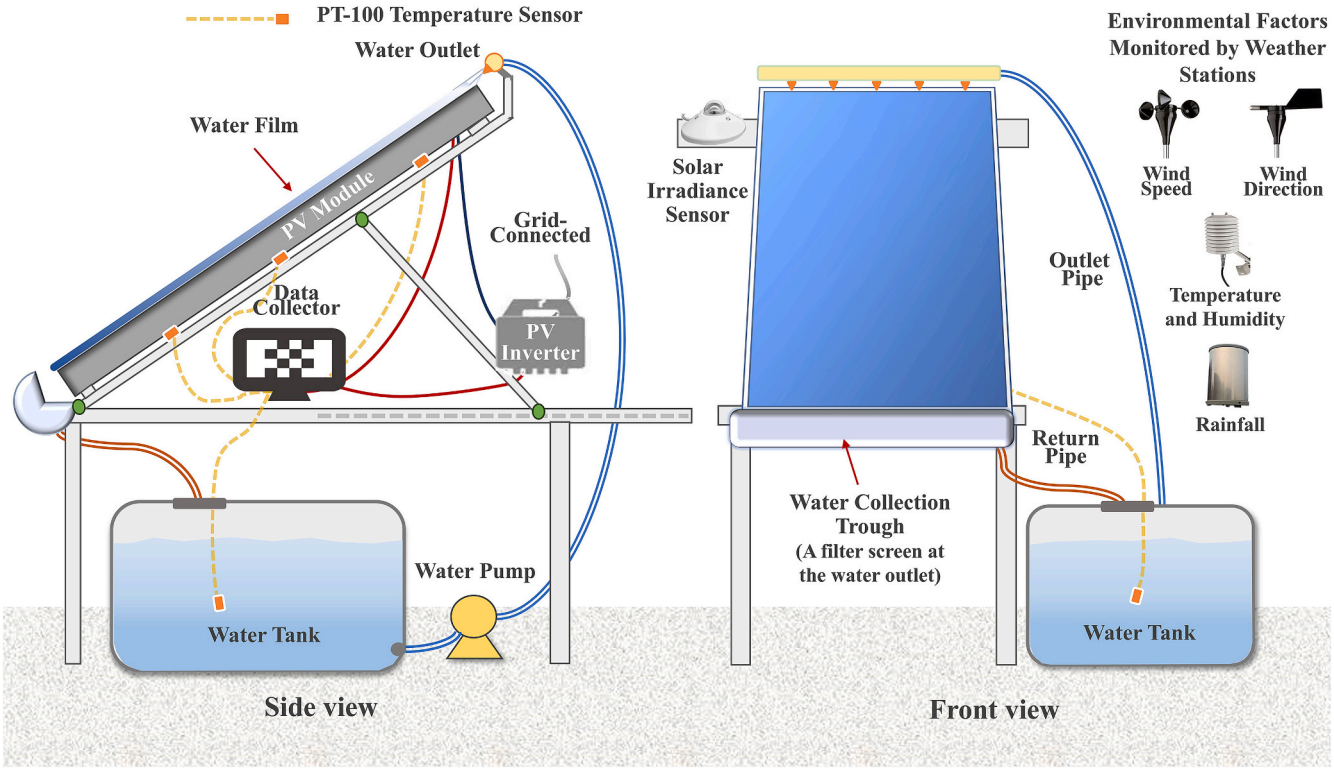


Fig. 1. Schematic diagram of active cooling system for photovoltaic modules.

thermal–electrical model is developed for the photovoltaic module with active water cooling. The model captures the transient energy transfer within the module's multilayer structure as well as the heat exchange associated with the active water-cooling process. It establishes a coupling between the thermal and electrical behaviors of the module. Additionally, the model accounts for variations in cooling water volume, including water losses during operation, and their impact on thermal regulation and power output.

2.1. Thermal model development for a photovoltaic module with active water-cooling

In this thermal model, the metal frame of the photovoltaic module is not modeled, as its surface area is relatively small compared to the total module area, and its impact on the temperature response is negligible [33]. For the thermal model of the photovoltaic module with a water-cooling system, the following assumptions are made [34,35]:

(i) Energy transfer is one-dimensional and time-varying;
(ii) Thermal conduction is assumed to be uniform across the length and width of the photovoltaic module;
(iii) The temperature at the center of each layer of the photovoltaic module represents the average temperature of that layer;
(iv) There is no contact thermal resistance between the layers inside the module;
(v) Heat transfer does not occur on the sides of the photovoltaic module;
(vi) The EVA layer does not absorb solar radiation.

To model the thermal interactions within the water-cooling loop, the model incorporates multiple heat exchange interfaces within the cooling loop, including: heat transfer between the water film and the module; thermal interaction between the returning water and the tank fluid; and heat dissipation from the tank water to the tank wall and subsequently to the ambient environment. It dynamically tracks the evolution of water temperature and volume in the reservoir, while accounting for key hydraulic parameters such as flow rate and film velocity. This closed-loop formulation captures the bidirectional coupling between fluid transport and thermal regulation, providing a robust basis for evaluating thermal performance and the resulting electrical output under varying cooling conditions.

Fig. 2 presents the equivalent thermal circuit of a photovoltaic module integrated with a water-cooling system. The thermal network describes the primary heat transfer pathways within the module and between the module and its surroundings. Thermal resistances represent conductive heat transfer through the multilayer structure of the module, including the glass layer, EVA1 layer, solar cell layer, EVA2 layer, and back sheet layer. Convective and radiative thermal resistances characterize the heat exchange between the module surfaces and the surrounding environment.

When the water-cooling system is activated, additional heat transfer pathways are introduced. These include convective heat transfer between the water film and the ambient air, radiative heat exchange between the water film surface and the sky, evaporative heat transfer from the water film surface, and thermal interaction between the circulating water film and the water stored in the tank, etc. The thermal resistances corresponding to these processes are indicated in Fig. 2, and their definitions are summarized in Table 1.

A mode selector switch (S) enables the transition between two operating states. When S = 1 (connected to position 1), the module operates without cooling condition, relying solely on ambient heat dissipation. When S = 2 (connected to position 2), active water cooling is engaged, and a thin, continuous water film forms over the front surface of the module to enhance thermal regulation.

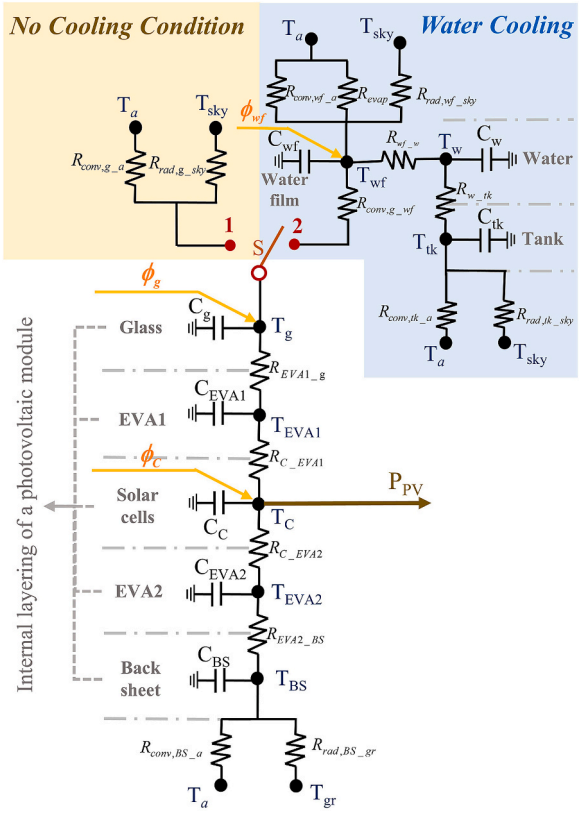


Fig. 2. The equivalent thermal circuit of photovoltaic modules with a water-cooling system.

Table 1
Definition of thermal resistances used in the equivalent thermal circuit.

Thermal resistance	Description
R_{conv,tk_a}	Convection thermal resistance between the tank body and the ambient
R_{rad,tk_sky}	Radiation thermal resistance between the tank body and the sky
R_{w_tk}	Conduction thermal resistance between the water in the box and the tank body
R_{wf_w}	Thermal resistance between the water film and the water in the tank
R_{conv,g_wf}	Convection thermal resistance between the glass layer and the water film
R_{conv,wf_a}	Convection thermal resistance between the water film and the ambient
R_{rad,wf_sky}	Radiation thermal resistance between the water film and the sky
R_{evap}	Evaporation thermal resistance of the water film
R_{conv,g_a}	Convection thermal resistance between the glass layer and the ambient
R_{rad,g_sky}	Radiation thermal resistance between the glass layer and the sky
R_{EVA1_g}	Conduction thermal resistance between the EVA1 layer and the glass layer
R_{C_EVA1}	Conduction thermal resistance between the solar cells layer and the EVA1 layer
R_{C_EVA2}	Conduction thermal resistance between the solar cells layer and the EVA2 layer
R_{EVA2_BS}	Conduction thermal resistance between the EVA2 layer and the back sheet layer
R_{conv,BS_a}	Convection thermal resistance between the back sheet layer and the ambient
R_{rad,BS_gr}	Radiation thermal resistance between the back sheet layer and the roof surface

2.1.1. Formulation of multi-node energy balance equations with water-cooling integration

This section presents the multi-node heat transfer equations used in

the development of the thermal model for the photovoltaic module with a water-cooling system, including two operating states: no cooling and active water cooling. In these equations, the subscripts " tk " " wf " " g " " EVA " " C " " BS " " gr " " w " and " a "denote the water tank body, the water film, the glass layer, the EVA layer, the solar cells layer, the back sheet layer, the roof surface, the water in the tank and ambient. According to the energy transfer principle, the relevant heat transfer equation in the thermal model is expressed as follows:

(1) For the water tank body

$$c_{p,tk}\delta_{tk}A_{tk}\rho_{tk}\frac{dT_{tk}}{dt}=\frac{T_w-T_{tk}}{R_{w_tk}}-\frac{T_{tk}-T_a}{R_{conv,tk_a}}-\frac{T_{tk}-T_{sky}}{R_{rad,tk_sky}} \tag{1}$$

where c_p is the specific heat,δ is the thickness,ρ is the density,A_{tk} is the surface area of the water tank, t is the time, T is the temperature, T_{sky} is the sky temperature, which can be expressed as [34]:

$$T_{sky}=0.0552T_a^{1.5} \tag{2}$$

(2) For the water in the tank

$$\begin{cases} c_{p,w}m_w(t)\dfrac{dT_w}{dt}=\dfrac{T_{wf}-T_w}{R_{wf_w}}-\dfrac{T_w-T_{tk}}{R_{w_tk}} \\ m_w(t)=m_0-\displaystyle\int_0^t \dot{m}_{loss}(\psi)d\psi \end{cases} \tag{3}$$

where $m_w(t)$ the mass of water in the tank at time t, m_0 is the initial water mass in the water tank, $\dot{m}_{loss}$ is the mass flow rate of water loss during water cooling process.

(3) For the water film

$$c_{p,w}\delta_{wf}A\rho_w\frac{dT_{wf}}{dt}=\alpha_{wf}GA+\frac{T_g-T_{wf}}{R_{conv,g_wf}}-\frac{T_{wf}-T_a}{R_{conv,wf_a}}-\frac{T_{wf}-T_{sky}}{R_{rad,wf_sky}}-\frac{T_{wf}-T_a}{R_{evap}}-\frac{T_{wf}-T_w}{R_{wf_w}} \tag{4}$$

where A is the area of the photovoltaic module (A=$H\times W$), H and W are the length and width of the photovoltaic module, respectively. α is the absorptivity, G is the incident solar radiation on the module plane.

(4) For the glass layer

$$c_{p,g}\delta_g A\rho_g\frac{dT_g}{dt}=\alpha_g\tau_{wf}GA+\frac{T_{EVA1}-T_g}{R_{EVA1_g}}-\frac{T_g-T_{wf}}{R_{conv,g_wf}} \tag{5}$$

For the energy balance of the glass layer under no cooling conditions:

$$c_{p,g}\delta_g A\rho_g\frac{dT_g}{dt}=\alpha_g GA+\frac{T_{EVA1}-T_g}{R_{EVA1_g}}-\frac{T_g-T_a}{R_{conv,g_a}}-\frac{T_g-T_{sky}}{R_{rad,g_sky}} \tag{6}$$

where τ is the transmissivity.

(5) For the EVA1 layer

$$c_{p,EVA}\delta_{EVA}A\rho_{EVA}\frac{dT_{EVA1}}{dt}=\frac{T_C-T_{EVA1}}{R_{C_EVA1}}-\frac{T_{EVA1}-T_g}{R_{EVA1_g}} \tag{7}$$

The energy balance of the EVA1 layer under the no cooling condition is the same as Eq. (7).

(6) For the solar cells layer

$$c_{p,C}\delta_C A\rho_C\frac{dT_C}{dt}=\alpha_C\tau_g\tau_{wf}GA-P-\frac{T_C-T_{EVA1}}{R_{C_EVA1}}-\frac{T_C-T_{EVA2}}{R_{C_EVA2}} \tag{8}$$

For the energy balance of the solar cells layer under no cooling conditions:

$$c_{p,C}\delta_C A\rho_C\frac{dT_C}{dt}=\alpha_C\tau_g GA-P-\frac{T_C-T_{EVA1}}{R_{C_EVA1}}-\frac{T_C-T_{EVA2}}{R_{C_EVA2}} \tag{9}$$

where P is the power of the photovoltaic module.

(7) For the EVA2 layer

$$c_{p,EVA}\delta_{EVA}A\rho_{EVA}\frac{dT_{EVA2}}{dt}=\frac{T_C-T_{EVA2}}{R_{C_EVA2}}-\frac{T_{EVA2}-T_{BS}}{R_{EVA2_BS}} \tag{10}$$

The energy balance of the EVA2 layer under the no cooling condition is the same as Eq. (10).

(8) For the back sheet layer

$$c_{p,BS}\delta_{BS}A\rho_{BS}\frac{dT_{BS}}{dt}=\frac{T_{EVA2}-T_{BS}}{R_{EVA2_BS}}-\frac{T_{BS}-T_a}{R_{conv,BS_a}}-\frac{T_{BS}-T_{gr}}{R_{rad,BS_gr}} \tag{11}$$

The energy balance of the back sheet layer under the no cooling condition is the same as Eq. (11).

2.1.2. Thermal resistance and auxiliary parameters

Thermal resistance is a critical parameter in thermal management. In the thermal model of the photovoltaic module with water-cooled system, thermal resistance plays a crucial role in the heat transfer energy within the system and between the system and the surrounding environment. Depending on the mode of heat transfer, the thermal resistance in heat transfer equations from Section 2.1.1 can be classified into four main types: radiation thermal resistance, convection thermal resistance, conduction thermal resistance, and evaporative thermal resistance.

For the radiation thermal resistance of the system, the energy dissipated by radiation is between the system and the environment. Therefore, the radiation thermal resistance R_{rad,tk_sky}, R_{rad,wf_sky}, R_{rad,g_sky} and R_{rad,BS_gr} of the water tank body, the water film layer, the glass layer and the back sheet layer can be expressed as [21,36]:

$$R_{rad,tk_sky}=\frac{1}{h_{rad,tk_sky}A_{tk}}=\frac{1}{\sigma\varepsilon_1\left(T_{tk}^2+T_{sky}^2\right)\left(T_{tk}+T_{sky}\right)A_{tk}} \tag{12}$$

$$R_{rad,wf_sky}=\frac{1}{h_{rad,wf_sky}A}=\frac{1}{\sigma\varepsilon_2\left(T_{wf}^2+T_{sky}^2\right)\left(T_{wf}+T_{sky}\right)A} \tag{13}$$

$$R_{rad,g_sky}=\frac{1}{h_{rad,g_sky}A}=\frac{1}{\sigma\varepsilon_3\left(T_g^2+T_{sky}^2\right)\left(T_g+T_{sky}\right)A} \tag{14}$$

$$R_{rad,BS_gr}=\frac{1}{h_{rad,BS_gr}A}=\frac{1}{\sigma\varepsilon_4\left(T_{BS}^2+T_{gr}^2\right)\left(T_{BS}+T_{gr}\right)A} \tag{15}$$

where h_{rad,tk_sky}, h_{rad,wf_sky}, h_{rad,g_sky} and h_{rad,BS_gr} are radiative heat transfer coefficients of the water tank body, the module glass layer and the back sheet layer, σ is the Stefan-Boltzmann constant (5.67×10^{-8} W/(m^2·K^4)), ε is the emissivity, including $\varepsilon_1=1/\left(1/\varepsilon_{tk}+1/\varepsilon_{sky}-1\right)$, $\varepsilon_2=1/\left(1/\varepsilon_w+1/\varepsilon_{sky}-1\right)$,$\varepsilon_3=1/\left(1/\varepsilon_g+1/\varepsilon_{sky}-1\right)$, and $\varepsilon_4=1/\left(1/\varepsilon_{BS}+1/\varepsilon_{gr}-1\right)$.

In the photovoltaic module with a water-cooling system, the convective thermal resistance in the heat transfer process is categorized into two types: one for air as the fluid and another for water as the fluid. The convection thermal resistance related to air includes: R_{conv,tk_a}, R_{conv,wf_a},R_{conv,g_a} and R_{conv,BS_a}. The convection thermal resistance related to water includes: R_{conv,g_wf}. The calculation of convection thermal resistance is as follows [37]:

$$R_{conv}=\frac{1}{h_{conv}A'} \tag{16}$$

$$h_{conv}=\left(h_{conv,free}^3+h_{conv,forced}^3\right)^{\frac{1}{3}} \tag{17}$$

$$h_{conv,\mathrm{free/\ forced}}=\frac{k\cdot Nu_{\mathrm{free/\ forced}}}{L} \tag{18}$$

where h_{conv} is the convection heat transfer coefficient, A' is the contact area with the fluid. For R_{conv,tk_a}, the contact area is A_{tk}, and for R_{conv,wf_a}, R_{conv,BS_a}, R_{conv,g_a}, R_{conv,g_wf}, the contact area is A. k is the thermal

conductivity of air/water. L is the characteristic length. Nu is the Nusselt number. Due to the different driving forces of the flow, the Nusselt number is calculated differently in natural convection and forced convection. The two are expressed as follows [35,37]:

$$Nu_{free} = f(Pr, Ra) \tag{19}$$

$$Nu_{forced} = f(Pr, Re) \tag{20}$$

where Pr, Ra, and Re are Prandtl number, Reynolds number, and Rayleigh number, respectively. These parameters vary with the material properties and the geometry of the object.

$$Pr = \frac{\upsilon}{\alpha_d} \tag{21}$$

$$Ra = Gr{\cdot}Pr \tag{22}$$

$$Gr = \frac{g{\cdot}\beta{\cdot}\Delta T{\cdot}L^3}{\upsilon^2} \tag{23}$$

$$Re = \frac{v{\cdot}L}{\upsilon} \tag{24}$$

where υ, α_d, and β are the kinematic viscosity, thermal diffusivity, and coefficient of isobaric volume expansion, with values depending on the specific fluid (e.g., water or air). v is the fluid velocity including wind speed and water flow speed, ΔT is the temperature difference. g is the gravity acceleration. It is worth noting that for the back sheet layer, $g{\cdot}sin\theta$ should be used instead of g in the Grashof number. θ is the tilt angle between the photovoltaic module and the horizontal plane.

According to the heat exchange conditions of different surfaces and media, a suitable Nusselt number calculation model is required to accurately describe the thermal convection characteristics of each area. For natural convection in the thermal model of the photovoltaic module with water-cooled system, Nu_{free} can be calculated as follows:

For the water tank body [37,38]:

$$Nu_{free} = 0.59Ra^{\frac{1}{4}}(10^4 \le Ra < 10^9) \tag{25}$$

$$Nu_{free} = 0.10Ra^{\frac{1}{3}}(10^9 \le Ra < 10^{13}) \tag{26}$$

For the glass layer and water film layer:

$$Nu_{free} = 0.54Ra^{\frac{1}{4}}(10^4 \le Ra \le 10^7) \tag{27}$$

$$Nu_{free} = 0.15Ra^{\frac{1}{3}}(10^7 < Ra \le 10^{11}) \tag{28}$$

For the back sheet layer:

$$Nu_{free} = 0.56Ra^{\frac{1}{4}}(10^5 < Ra < 10^{11}) \tag{29}$$

For the water film, its evaporative heat dissipation reduces the temperature difference between the water film and the air, weakening buoyancy-driven natural convection, making its effect negligible.

Moreover, the Nusselt number Nu_{forced} for forced convection in the thermal model can also be calculated as follows:

For the water tank body, water film layer, glass layer and back sheet layer [37,38]:

$$Nu_{forced} = \frac{0.3387Re^{\frac{1}{2}}Pr^{\frac{1}{3}}}{\left[1 + (0.0468/Pr)^{\frac{2}{3}}\right]^{\frac{1}{4}}}\left(Re \le 5 \times 10^5\right) \tag{30}$$

$$Nu_{forced} = 0.037Re^{\frac{4}{5}}Pr^{\frac{1}{3}}\left(0.6 < Pr < 60, Re > 5 \times 10^5\right) \tag{31}$$

For the glass layer with an active water-cooling, when $Re > 5 \times 10^5$, Nu_{forced} is calculated using Eq. (31). Conversely, Nu_{forced} is determined using the following equation [39,40]:

$$Nu_{forced} = 0.664Re^{\frac{1}{2}}Pr^{\frac{1}{3}}\left(Re \le 5 \times 10^5\right) \tag{32}$$

Evaporative thermal resistance is primarily associated with the heat transfer process of liquid evaporation and is commonly found in cooling systems. It can be obtained as Eq. (33):

$$R_{evap} = \frac{T_{wf} - T_a}{\dot{m}''_{evap}L_vA} \tag{33}$$

where L_v is the latent heat of vaporization, $\dot{m}''_{evap}$ is the evaporation mass flux per unit area. It can be represented as:

$$\dot{m}''_{evap} = h_m\left(\rho_{v,w} - \frac{RH}{100}\rho_{v,a}\right) \tag{34}$$

$$\rho_{v,w} = \frac{p_w}{R_vT_{wf}} \tag{35}$$

$$\rho_{v,a} = \frac{p_a}{R_vT_a} \tag{36}$$

where $\rho_{v,a}$ is the saturated vapor densities at the ambient temperature,$\rho_{v,w}$ is the saturated vapor densities at the water-film surface,p_a is the saturated vapor pressure at the ambient temperature,p_w is the saturated vapor pressure at the water-film surface, R_v is the gas constant for water vapor (461.5 J/(kg·K)), h_m is the mass transfer coefficient, calculated using the Sherwood number (Sh) correlation, expressed as [41]:

$$h_m = \frac{Sh{\cdot}D_v}{L} \tag{37}$$

$$Sh = 0.664Re^{\frac{1}{2}}Sc^{\frac{1}{3}} \tag{38}$$

$$Sc = \frac{\upsilon_a}{D_v} \tag{39}$$

where D_v is the diffusion coefficient of water vapor in air, Sc is the Schmidt number.

For conductive thermal resistance between adjacent nodes of the system, which all can be calculated as:

$$R_{i_j} = \frac{\delta_i}{2k_iA_i} + \frac{\delta_j}{2k_jA_j} \tag{40}$$

where the subscript 'i' and 'j' refer to the node i, j, involving water tank body, water body, water film, glass, EVA1, solar cells, EVA2, and back sheet, R_{ij} is the conductive thermal resistance between the node i and j.

The heat carried away by the cooling water during the cooling process is calculated as follows [21]:

$$\frac{T_{wf} - T_w}{R_{wf_w}} = \dot{m}''_{out}c_{p,w}\left(T_{wf} - T_w\right)A \tag{41}$$

$$\frac{d\left(\rho_w\delta_{wf}\right)}{dt} = \dot{m}''_{in} - \dot{m}''_{evap} - \dot{m}''_{out} \tag{42}$$

where $\dot{m}''_{out}$ is the mass flow rate of overflow water per unit area. $\dot{m}''_{in}$ is the mass flow rate of the supplied cooling water per unit area. δ_{wf} is the thickness of the water film, which can be calculated $Q_F/(v_{wf}{\cdot}W)$. Q_F is the water flow rate, v_{wf} is the water film velocity, which is determined based on a classical thin-film flow model derived from the governing momentum equation. The equation is as follows [42,43]:

$$v_{wf} = \frac{\delta_{wf}^2{\cdot}\rho_w{\cdot}g{\cdot}sin\theta}{3\mu} = \sqrt[3]{\frac{\rho_w{\cdot}g{\cdot}sin\theta{\cdot}Q_F^2}{3\mu W^2}} \tag{43}$$

where ρ_w is the density of water, μ is the dynamic viscosity.

2.2. Electrical modeling and solution of the coupled model

In the calculation of the thermal model in Section 2.1, the photovoltaic module power generation power P is calculated using Eq. (44) in the electrical model.

$$P = I_{m_new} \times U_{m_new} \tag{44}$$

where I_{m_new} and Um_new are the actual working voltage and current of the photovoltaic module corrected according to the influence of irradiance and photovoltaic module temperature. The two parameters can be obtained from the following photovoltaic module equivalent circuit model [44,45]:

$$I = I_{sc}\left[1 - C_1\left(\exp\left(\frac{U}{C_2 U_{oc}}\right) - 1\right)\right] \tag{45}$$

$$C_1 = \left(1 - \frac{I_m}{I_{sc}}\right)\exp\left(\frac{-U_m}{C_2 U_{oc}}\right) \tag{46}$$

$$C_2 = \left(\frac{U_m}{U_{oc}} - 1\right)/\ln\left(1 - \frac{I_m}{I_{sc}}\right) \tag{47}$$

where I_{sc} is the short circuit current, I_m is the current at the maximum power point, U_{oc} is the open circuit voltage, U_m and is the voltage at the maximum power point. The coefficients C_1 and C_2 can be derived from the nameplate parameters of the photovoltaic module, which are typically characterized under standard test conditions:

$$I_{sc_new} = I_{sc}\frac{G}{G_{ref}}\left[1 + \xi_I\left(T_C - T_{ref}\right)\right] \tag{48}$$

$$I_{m_new} = I_m\frac{G}{G_{ref}}\left[1 + \xi_I\left(T_C - T_{ref}\right)\right] \tag{49}$$

$$U_{oc_new} = U_{oc}\left[1 - \xi_U\left(T_C - T_{ref}\right)\right]\ln\left[e + \xi\left(\frac{G}{G_{ref}} - 1\right)\right] \tag{50}$$

$$U_{m_new} = U_m\left[1 - \xi_U\left(T_C - T_{ref}\right)\right]\ln\left[e + \xi\left(\frac{G}{G_{ref}} - 1\right)\right] \tag{51}$$

where ξ is the compensation coefficient, ξ_I is the current temperature coefficient, ξ_U is the voltage temperature coefficient.

The coupled numerical model was formulated and executed within the MATLAB R2018a computational environment. Key input parameters encompass the photovoltaic module's geometric configuration, thermophysical material properties, installation inclination angle, and ambient operating conditions. The governing equations describing the thermal behavior of the photovoltaic module consist of a set of transient energy balance equations for each stratified layer. These equations form a system of ordinary differential equations (ODEs) that is solved using a time-marching numerical scheme. The simulation begins with the initialization of the temperature profile across each stratified layer of the photovoltaic module. Subsequently, the electrical output is determined using the electrical sub-model based on the current temperature state, from which the internal heat generation is calculated. The thermal sub-model then updates the temperature field of each layer according to the corresponding energy balance equations.

In the numerical solution procedure, the thermal and electrical sub-models are sequentially coupled at each time step, and an iterative scheme is adopted to ensure convergence of the coupled thermal–electrical solution. After each iteration, the updated temperatures overwrite the previous values and are used for the next calculation cycle. The iterative process continues until dual termination criteria are satisfied: 1) The maximum relative error between successive iterations is less than a tolerance threshold of 1×10^{-5}, and 2) The simulation duration reaches the prescribed temporal boundary. At this point, the final temperature distribution and the electrical output power of the photovoltaic module are obtained, allowing the thermal–electrical behavior of the photovoltaic module to be captured under varying operating conditions.

2.3. Performance indicators

The system analysis accounts for the electrical consumption of the circulation pump, which represents an auxiliary energy cost that partially offsets the power gain provided by the cooling system. The primary objective of integrating an active water-cooling system into photovoltaic modules is to reduce their operating temperature and thereby enhance power output. To evaluate the performance enhancement provided by such cooling systems in a way that is both generalizable and independent of module size or configuration, a dimensionless performance indicator, the net power gain ratio (ΔP_{rat}), is introduced. This metric enables a standardized assessment of the electrical benefits of active cooling [46]. Eq. (52) mathematically define this indicator based on simulation data.

$$\Delta P_{rat} = \frac{P_{WCC} - P_{pump} - P_{NCC}}{P_{NCC}} \times 100\% = \frac{P_{rat}}{P_{NCC}} \times 100\% \tag{52}$$

where P_{rat} is the net power gain of the photovoltaic module. P_{pump} is the power consumption of the water pump, which can be calculated by the Eq. (53) [20,47].

$$P_{pump} = \frac{Q_F \cdot Z_{total} \cdot g \cdot \rho_w}{\eta_{pump}} \tag{53}$$

$$Z_{total} = Z_0 + Z_{\text{loss}} \tag{54}$$

$$Z_{\text{loss}} = \frac{C_k \cdot L_p}{C_{hw}^{1.852} \cdot D^{4.87}} \cdot Q_F^n \tag{55}$$

where η_{pump} is the efficiency of the water pump, taken as 60%. Z_0 is the height difference between the water outlet and the water pump, which is 2 m in the study. Z_{loss} is the head loss through the water pipes, L_p is the length of the pipe, n is 1.852, C_{hw} is 130, and C_k is 10.67.

Water flow rate (WFR) is a key parameter governing the performance of the cooling system. Although higher flow rates generally enhance the cooling effect, they also increase the pumping power consumption, which does not necessarily lead to a proper P_{rat} [20]. Therefore, it is essential to identify the optimal water flow rate (OWFR) that maximizes the net power gain rate (ΔP_{rat_max}).

In practical outdoor photovoltaic systems, environmental parameters such as solar irradiance, ambient temperature, and wind speed vary continuously over time, making it difficult to systematically adjust the water flow rate while maintaining identical boundary conditions during field experiments. Consequently, a comprehensive experimental exploration of the optimal flow rate would require a large number of repeated tests under strictly controlled environmental conditions, which is difficult to achieve for real-world operating photovoltaic systems. Therefore, this study uses the thermal-electrical coupled model that has been developed and experimentally verified to calculate the optimal flow rate through model simulation optimization.

Based on the validated model, the water flow rate is varied within a predefined range, and the corresponding net power gain rate is calculated for each operating condition. This indicator accounts for both the increase in photovoltaic power output due to cooling and the additional pumping power consumption. By comparing the calculated results under different flow conditions, the flow rate corresponding to the maximum value of ΔP_{rat} is identified as the OWFR. It should be emphasized that the OWFR is determined individually for each set of environmental conditions through the model-based optimization process, reflecting the optimal operating state corresponding to specific operating scenarios. Fig. 3 illustrates the calculation process for determining OWFR.

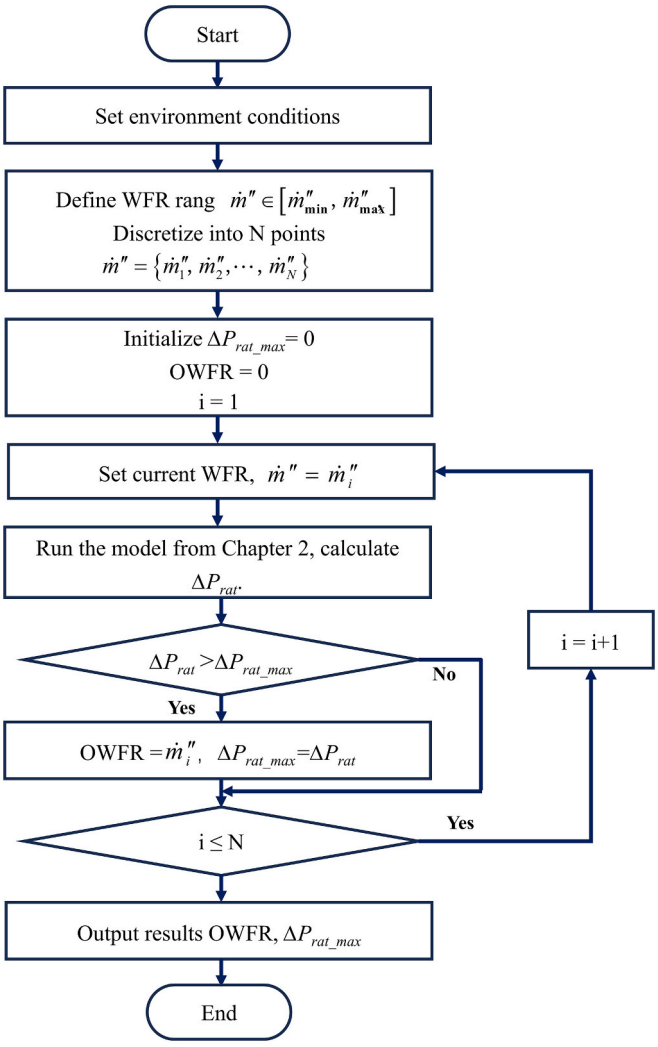


Fig. 3. Flowchart for identifying optimal operating conditions.

3. Description of the experiment

This section describes the experimental design of an active cooling system for photovoltaic modules with rainwater harvesting function located on the roof of a building. The data collected from the experiment is used for the analysis of the rainwater harvesting characteristics of the module and the verification of the model developed in this paper.

3.1. Construction of the experimental device

The photovoltaic test site is located on the roof of the Key Laboratory Building at Tianjin University, Tianjin, China, as shown in Fig. 4. The site is situated at 38.9942°N, 117.3146°E, within a typical temperate monsoon climate zone of northern China. The composition of the photovoltaic module active cooling system has been described previously, and its structural schematic is presented in Fig. 1. The electrical parameters of the photovoltaic module are summarized in Table 2. The active cooling system is designed to collect rainwater during rainfall events and to utilize the collected rainwater to provide water cooling for the photovoltaic modules under conditions of high irradiance and elevated temperatures.

The monitoring equipment installed at the site includes environmental monitoring sensors, a PT100 temperature sensor, and photovoltaic power generation monitoring equipment. Detailed specifications of each type of monitoring device are summarized in Table 3. All sensors were connected to a centralized data acquisition system to ensure continuous monitoring of environmental conditions and system operating parameters during the experiment. The monitoring system allows the sampling interval to be set at the minute level, providing sufficient temporal resolution to accurately capture the variations of environmental conditions and photovoltaic module responses. The recorded parameters included solar irradiance, rainfall intensity, wind speed, wind direction, ambient temperature, relative humidity, photovoltaic module temperature, water temperature in the water tank, and the electrical output parameters of the photovoltaic module.

In addition to the automatically recorded environmental data, the rainwater harvesting data were compiled on a rainfall-event basis. Individual rainfall events were identified based on the rain gauge records, and the amount of rainwater collected during each event was manually measured and recorded. This event-based measurement approach enabled accurate quantification of the harvested rainwater volume corresponding to each rainfall process.

The collected raw data were subsequently screened to eliminate abnormal values caused by temporary sensor disturbances or communication interruptions. The validated dataset was then organized and used for the analysis of rainwater harvesting characteristics and for the validation of the coupled model developed in this study.

3.2. Experimental design

The experiment consists of two parts: a rainfall harvesting experiment during the rainy season at different tilt angles of rooftop photovoltaic systems, and a water-cooling test of photovoltaic modules for model validation. The rainwater harvesting experiment was conducted during the local rainy season (from June to September) in both 2023 and 2024. The photovoltaic module azimuth of the photovoltaic module faced south, and the layout tilt angle in the experiment were set at 5°, 15°, 25°, 35°, and 45° (as shown in Fig. 3). Among these configurations, 35° lies within the recommended optimal tilt angle range for photovoltaic installations in Tianjin, which is close to the local latitude and is widely adopted in PV system design to maximize annual solar energy reception [48–50]. During rainfall, the surface of the photovoltaic modules at each tilt angle collects rainwater, which is then directed into a water tank via the trough and return pipe. Various types of sensors record environmental data, such as rain intensity, rainfall volume, wind speed, and wind direction, throughout the rainfall process. The amount of rainwater collected in each rainfall event is measured and tallied. The rainwater harvesting efficiency (η_{rain}) of each module layout is evaluated using the following equation:

$$\eta_{rain} = \frac{V_{rain}}{H_R \cdot A_E} \times 100\% \tag{56}$$

where V_{rain} is the volume of rainwater harvesting, H_R is the rainfall, A_E is the effective catchment area.

The photovoltaic module water cooling test is carried out on sunny days when the modules are at high temperatures. To be more representative, the test focused on photovoltaic modules arranged according to the optimal radiation tilt angle. Rainwater stored in a water tank is pumped out by a water pump and flows through an outlet pipe to the surface of the photovoltaic module, with the flow rate controlled by the pump. The water flows directly over the module surface, forming a thin water film. Except for a small amount of water lost through evaporation and overflow, the remaining water is collected in a trough and returned to the water tank. During the experiment, the temperature of the photovoltaic module, the water temperature in the tank, and the electrical power generation parameters were continuously recorded by the data acquisition system at 1 min intervals.

It should be noted that the evaporation rate of the water film was not directly measured during the experiment. Instead, it was estimated within the coupled model using a unit-area evaporation rate correlation



Fig. 4. Photovoltaic device test site.

Table 2
Electrical parameters of the photovoltaic module.

Description	Symbol	Value
Maximum power	P	280 W(0~+5 W)
Maximum power point voltage	U_m	31.95 V
Maximum power point current	I_m	8.77 A
Open circuit voltage	U_{oc}	39.22 V
Short circuit current	I_{sc}	9.31 A

Table 3
Equipment instrument parameters.

Type (sensor)	Model of the equipment	Measuring accuracy	Measuring limit
Solar radiation	TBQ-2	< 5%	0 to 2000 W/m^2
Rain gauge	PC-2Y	0.1 mm	0 to 999.9 mm
Wind speed	PC-4GF	±0.3 m/s	0 to 70 m/s
Wind direction		±3°	0 to 360°
Ambient temperature		±0.2 ℃	−40 to 80 ℃
Relative humidity		±2%	0 to 100% RH
Photovoltaic module temperature and water temperature	PT100	0.15 ℃	−50 to 180 ℃
Power generation parameters for photovoltaic	TP1000	≤ 0.5%	0 to 120 V 0 to 20 A

described in Section 2.1.2, and the calculation results were verified by measuring the change in cooling water volume through water-cooling experiments.

3.3. Uncertainty analysis

To verify the reliability of experimental data, a systematic uncertainty analysis was conducted for all monitoring instruments and derived parameters, following the propagation of uncertainty method. The standard uncertainty (u) of each measuring instrument was calculated based on its measurement accuracy (a) in Table 3, using the following formula [51,52]:

$$u = \frac{a}{\sqrt{3}} \tag{57}$$

For each measured parameter λ, the total uncertainty ($\delta\lambda$) consists of combining the repetition uncertainty ($\delta\lambda_{rep}$) and the standard instrument uncertainty ($\delta\lambda_{stp}$),with $\delta\lambda_{stp}$ calculated according to Eq. (57). The synthesis formula is expressed as follows:

$$\delta\lambda = \sqrt{\left(\delta\lambda_{rep}\right)^2 + \left(\delta\lambda_{stp}\right)^2} \tag{58}$$

For the key derived parameters, which are functions of multiple independent variables, the square root method is used to calculate the combined uncertainty. If F is a function of 'n' independent linear parameters as $F = f(\lambda_1, \lambda_2, \cdots, \lambda_n)$, the uncertainty of function F may be calculated as [53]:

$$\delta F = \sqrt{\left(\frac{\partial F}{\partial \lambda_1}\delta\lambda_1\right)^2 + \left(\frac{\partial F}{\partial \lambda_2}\delta\lambda_2\right)^2 + \cdots + \left(\frac{\partial F}{\partial \lambda_n}\delta\lambda_n\right)^2} \tag{59}$$

Uncertainty analysis was performed on the measured variable and derived parameters using the uncertainty propagation method based on the 95% confidence interval. For example, the relative uncertainties of

derived parameters P and η_{rain} are calculated using the following formulas:

$$\frac{\delta P}{P} = \sqrt{\left(\frac{\partial U}{U}\right)^2 + \left(\frac{\partial I}{I}\right)^2} \tag{60}$$

$$\frac{\delta \eta_{rain}}{\eta_{rain}} = \sqrt{\left(\frac{\partial V_{rain}}{V_{rain}}\right)^2 + \left(\frac{\partial H_R}{H_R}\right)^2 + \left(\frac{\partial A_E}{A_E}\right)^2} \tag{61}$$

The maximum relative uncertainties of the key experimental parameters were determined. The results show that the uncertainties of the photovoltaic power, rainwater harvesting efficiency, and photovoltaic module temperature are 1.47%, 2.22%, and 0.53%, respectively, indicating that the experimental measurements have acceptable accuracy and reliability.

4. Results and discussion

This section presents the results of experimental testing and modeling analysis. Section 4.1 validates the accuracy and reliability of the developed model through active water-cooling experiments on rooftop photovoltaic modules. Section 4.2 determines the influence of environmental factors on cooling-induced net power gain ratio under different water flow rates. Section 4.3 evaluates seasonal power gains and evaporative losses of photovoltaic modules under optimal cooling conditions. Finally, Section 4.4 explores rainwater harvesting by photovoltaic modules and highlights its implications for sustainable cooling water management.

4.1. Validation of the coupled model for photovoltaic modules with a water-cooling system

To evaluate the accuracy and reliability of the proposed dynamic thermal–electrical coupled model for photovoltaic modules, experimental data obtained from water-cooling tests were used for model validation. The validation experiment was conducted during clear-sky summer periods when the photovoltaic modules experienced relatively high solar irradiance and elevated operating temperatures. Under such conditions, thermal loads on the modules are significant, making them representative scenarios for assessing the performance of water-cooling systems. In these experiments, rainwater collected by the photovoltaic modules was used as the cooling medium. Fig. 5 presents the comparative test cases in which the photovoltaic module operated under two modes: without cooling and with active water-cooling. The model simulations were performed under the same environmental and operating conditions as those recorded during the experiment. Fig. 6 compares the simulated results with the measured data for both operating modes.

In the model validation process, the mean absolute error (MAE), root mean square error (RMSE), and normalized root mean square error (NRMSE) were employed as performance metrics to quantitatively evaluate the agreement between the model simulation and the measured data.

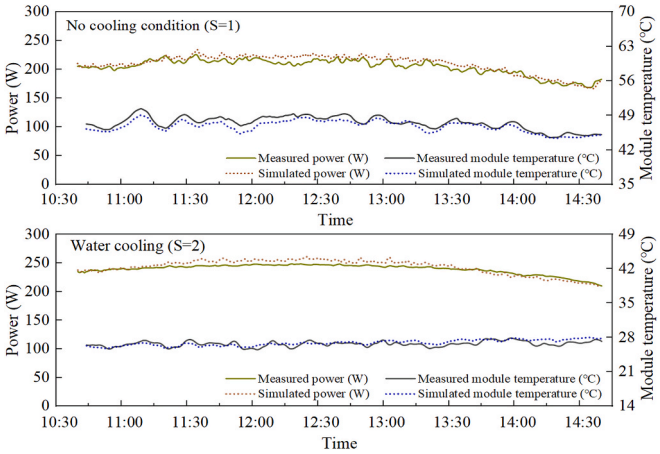


Fig. 6. Validation of a dynamic thermal–electrical coupled model for photovoltaic modules with an active water-cooling system.

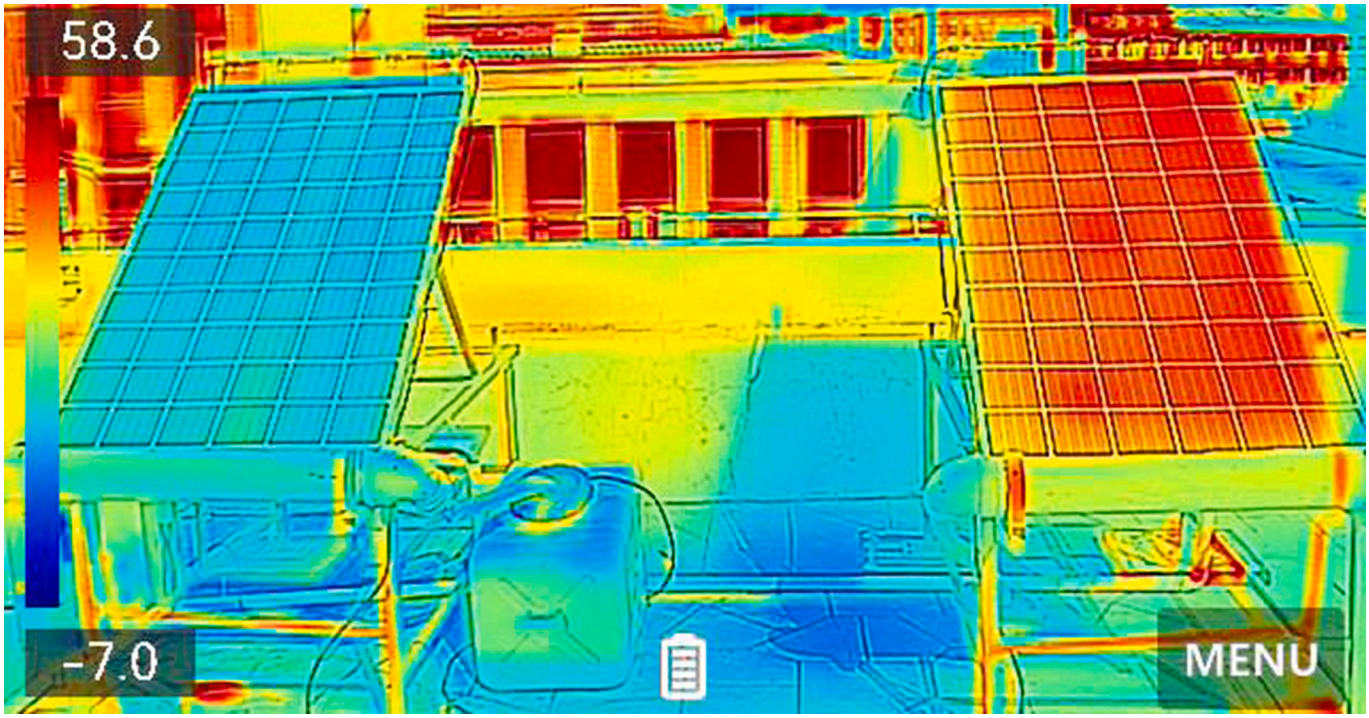


Fig. 5. Comparative test of active water-cooling of photovoltaic modules.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_{m,i} - Y_{c,i}\right| \tag{62}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_{m,i} - Y_{c,i}\right)^2} \tag{63}$$

$$NRMSE(\%) = \frac{RMSE}{\overline{Y}_m} \times 100\% \tag{64}$$

where Y_m is the measured value, Y_c is the simulated value.

Table 4 summarizes the evaluation results for the photovoltaic power output and module temperature under both no-cooling and water-cooling conditions. Under the no-cooling condition, the RMSE for power output was 7.767 W and the RMSE for module temperature was 1.115 ℃, with corresponding NRMSE values of 3.80% and 2.34%, respectively. Under the water-cooling condition, the RMSE values were reduced to 6.915 W for power output and 0.592 ℃ for module temperature, while the NRMSE values remained low at 2.89% and 2.21%. In all cases, the NRMSE values were lower than 5%, indicating a strong agreement between the simulated and measured data [54].

It should be explicitly noted that this model validation was conducted under specific experimental conditions. Nevertheless, the proposed physics-based dynamic thermal–electrical coupled model explicitly incorporates key environmental parameters as input variables, and its performance has been validated against the time-varying characteristics of these parameters under the tested conditions. In addition, this study employs a thin-film flow model to describe the effect of flow velocity, which has been shown in previous studies to reliably characterize water film behavior on photovoltaic module surfaces under a wide range of flow conditions [42]. Therefore, the favorable consistency between the simulated and measured results validates that the model can effectively characterize the thermal–electrical performance of photovoltaic modules under clear-sky conditions with high-temperature operation and water-cooling scenarios.

4.2. Influence of environmental factors on the performance of photovoltaic modules with a water-cooling system

Based on the validated thermal–electrical coupled model in Section 4.1, this section conducts a parametric analysis to investigate the influence of environmental factors on the active water-cooling performance of photovoltaic modules. Fig. 7 shows the response characteristics of cooling water flow to the net power gain ratio (ΔP_{rat}) of photovoltaic modules under different irradiance, ambient temperature and wind speed conditions. For all investigated environmental conditions, the ΔP_{rat} exhibits a non-monotonic variation with the water flow rate, increasing to a maximum and decreasing thereafter. This behavior indicates the existence of an OWFR at which the net power gain is maximized, resulting from the combined effects of enhanced thermal dissipation, diminishing marginal cooling benefit, and the associated increase in pumping power demand at higher flow rates.

Table 5 summarizes the ΔP_{rat_max} of modules obtained through active water cooling under different environmental combinations. Solar irradiance is identified as the dominant factor governing the achievable power gain. Under fixed ambient temperature and wind speed conditions, increasing irradiance from 800 to 1100 W/m^2 leads to an average relative increase of approximately 52% in the ΔP_{rat_max}. This pronounced enhancement is attributed to the substantially higher thermal load imposed on the photovoltaic module at elevated irradiance levels, which strengthens the effectiveness of active water cooling in suppressing temperature rise and thereby improving electrical performance [54]. Ambient temperature also exerts a significant influence on the net power gain achieved through active cooling. Under identical irradiance and wind speed conditions, higher ambient temperatures lead to a larger ΔP_{rat_max}. For instance, at a wind speed of 1 m/s and an irradiance of 1000 W/m^2, the ΔP_{rat_max} at 35 °C is 16.8% higher than that at 25 °C. This trend is associated with the higher operating temperature of the photovoltaic modules, which enhances the effectiveness of active water cooling.

In contrast, increasing wind speed results in a noticeable reduction in the ΔP_{rat_max}. Under identical irradiance and ambient temperature conditions, increasing the wind speed from 1 m/s to 3 m/s leads to a relative decrease in ΔP_{rat_max} ranging from 14.9% to 19.5%. The impact is most pronounced at 35 °C and 1100 W/m^2, where the wind increase causes the largest absolute reduction in water-cooling gain, corresponding to an absolute decrease 2.1 percentage points in ΔP_{rat_max}. This behavior can be attributed to the enhancement of natural convective heat transfer at higher wind speeds, which weakens the relative contribution of the water-cooling system. As a result, the incremental thermal and electrical benefits provided by active water cooling become less pronounced under strong wind conditions.

Overall, the combined results of Fig. 7 and Table 5 demonstrate that the effectiveness of the water-cooling system is strongly dependent on environmental conditions. Higher irradiance and ambient temperature favor greater power gain, whereas higher wind speed suppresses the achievable improvement. These findings highlight the necessity of considering coupled environmental factors when evaluating the performance enhancement potential of actively cooled photovoltaic modules.

Table 4
Model evaluation metrics.

Item		MAE	RMSE	NRMSE
NO cooling condition	Power	6.319 W	7.767 W	3.80%
	Module temperature	0.971 ℃	1.115 ℃	2.34%
Water cooling	Power	5.873 W	6.915 W	2.89%
	Module temperature	0.487 ℃	0.592 ℃	2.21%

4.3. Evaluation of water-cooling performance and associated evaporative losses

This section presents a seasonal case analysis to evaluate the module's improvement in power generation characteristics enabled by the cooling strategy under local operating conditions. The summer period from June to August represents a critical window characterized by high temperatures and intense solar radiation on sunny days and the resulting increased cooling requirements for photovoltaic modules. Its overlap with the regional rainy season—characteristic of monsoon-influenced regions during the summer months—provides a representative background for this analysis. Under such conditions, photovoltaic modules are subjected to intensified thermal loads, while the availability of rainwater offers practical support for the implementation of the proposed cooling strategy.

Fig. 8 shows the hourly averages of clear-sky meteorological parameters for each summer month in 2024. The data in the figure show the clear-sky day pattern in the temperate monsoon region in summer. Solar irradiance and ambient temperature peak around noon, corresponding to the period of maximum thermal stress on photovoltaic modules. Wind speed remains relatively stable throughout the day, while relative humidity reaches its minimum near midday. These meteorological conditions define the thermal boundary environment for evaluating the water-cooling characteristics of photovoltaic modules.

Fig. 9 illustrates the variations in net power gain (P_{rat}) of photovoltaic modules during a clear-sky summer day, calculated under the optimal water flow rate determined for each time period according to the corresponding environmental conditions. As shown, the P_{rat} exhibits a consistent diurnal pattern across all summer months, characterized by a gradual increase during the morning hours, a peak around midday, and a subsequent decline in the afternoon. Specifically, the maximum hourly

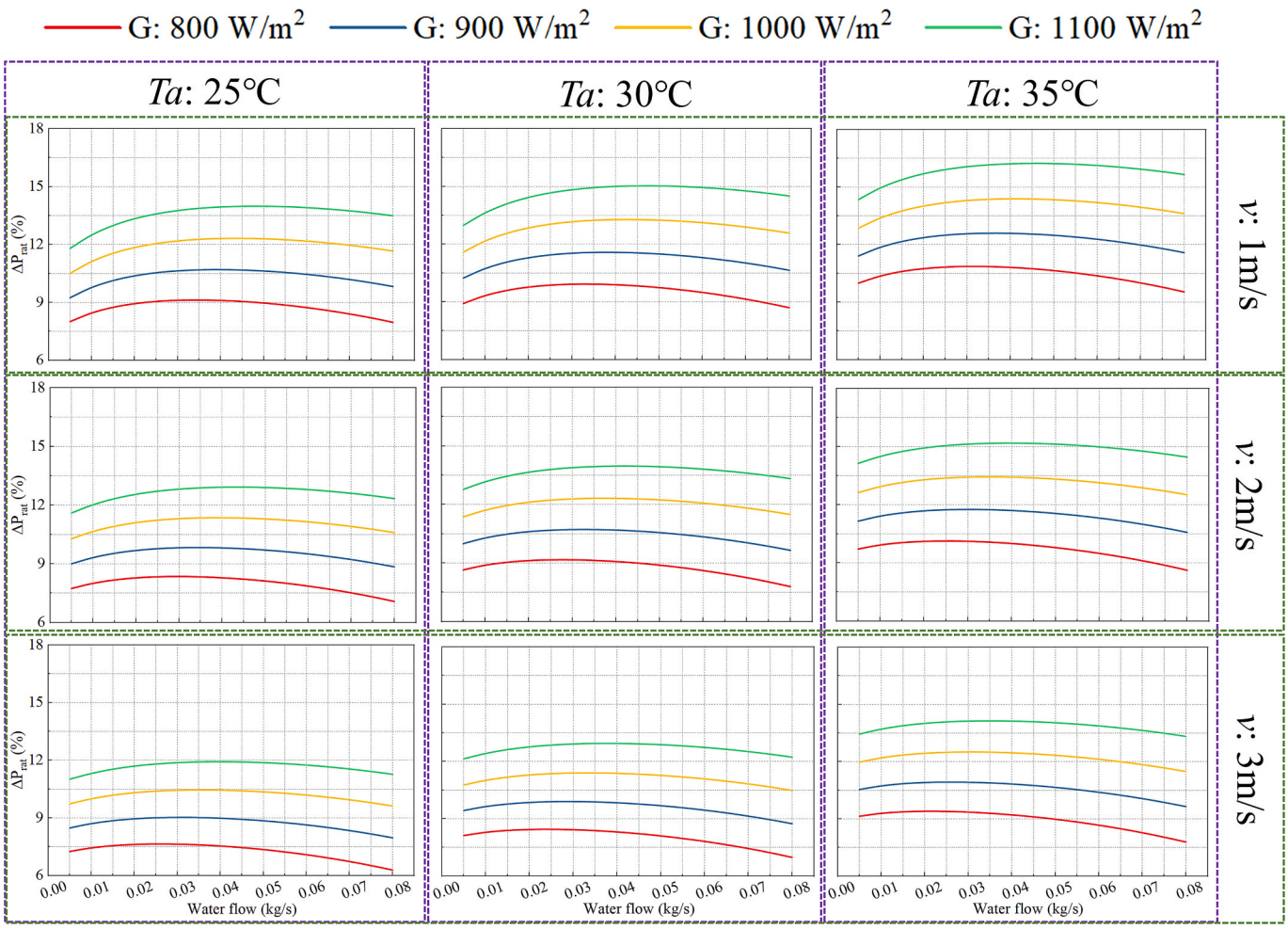


Fig. 7. Effect of water flow rate on ΔP_{rat} under different environmental conditions.

Table 5
The ΔP_{rat_max} under different environmental conditions.

Wind speed (m/s)	Ambient temperature (℃)	Irradiance (W/m²) 800	900	1000	1100
1	25	9.12%	10.69%	12.32%	13.98%
	30	9.92%	11.57%	13.28%	15.03%
	35	10.86%	12.59%	14.39%	16.24%
2	25	8.34%	9.83%	11.35%	12.92%
	30	9.16%	10.71%	12.32%	13.97%
	35	10.13%	11.75%	13.44%	15.18%
3	25	7.63%	9.02%	10.45%	11.92%
	30	8.44%	9.90%	11.40%	12.95%
	35	9.40%	10.93%	12.51%	14.14%

P_{rat} for each month consistently occurs between 11:00 and 12:00, exceeding 7.0% of the rated module capacity (i.e., based on the 280 W nominal power of each module). This indicates a substantial power enhancement due to water cooling under high-irradiance conditions, particularly around midday. In addition, the daily net energy gain of a single module under clear-sky conditions exceeds 100 Wh for all summer months, demonstrating that the water-cooling system provides a stable and sustained positive contribution to photovoltaic performance throughout the operating period.

A comparative analysis of the summer months further reveals that the overall level of P_{rat} in June is higher than that in July and August, with the daily net power generation gains increasing by approximately 13.8% and 27.7% relative to July and August, respectively. In addition, June also exhibited a broader plateau of high net gain around noon, indicating that the improvement in power generation performance was more sustained. This difference can be primarily attributed to the relatively higher solar irradiance levels and more favorable thermal environmental conditions in June, which amplify the cooling-induced performance enhancement.

The hourly clear-sky cooling water evaporation for each summer month under OWFR is shown in Fig. 10. In terms of trends, the evaporation rates in all months exhibit a consistent diurnal pattern, characterized by a pronounced single-peak distribution. The peak evaporation generally occurs within the period from 12:00 to 15:00, during which approximately 48–51% of the total daily evaporation takes place, indicating that water loss is strongly concentrated during the period of maximum solar radiation and thermal load on the photovoltaic modules. Furthermore, the ratio between the peak and minimum hourly evaporation rates ranges from approximately 2.5 to 3.3 across the three months, revealing a pronounced diurnal variability in evaporative demand under different meteorological conditions. Notably, the evaporation in June is significantly higher than in July and August, with the diurnal peak evaporation rate exceeding those of July and August by 42.9% and 97.7%, respectively. The corresponding daily water loss reaches 3.33 L/d in June, compared to 2.36 L/d and 1.64 L/d in July and August. This indicates that the total daily evaporation in June is approximately 41.1% higher than that in July and more than twice that in August. This discrepancy is primarily attributed to the higher ambient temperature, considerably lower relative humidity, and stronger wind

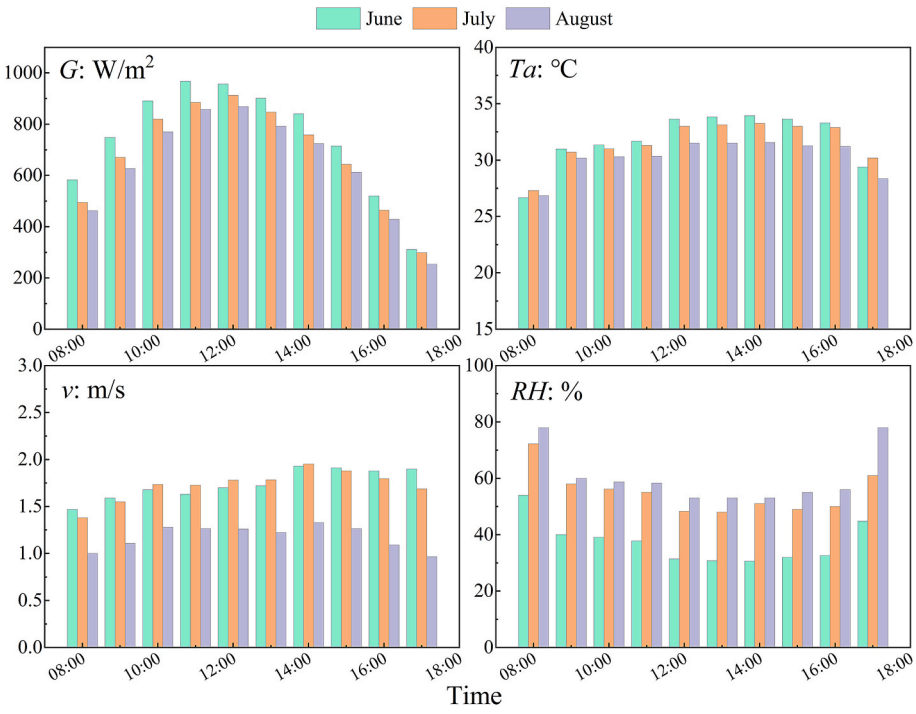


Fig. 8. Hourly average of clear-sky meteorological parameters within the month.

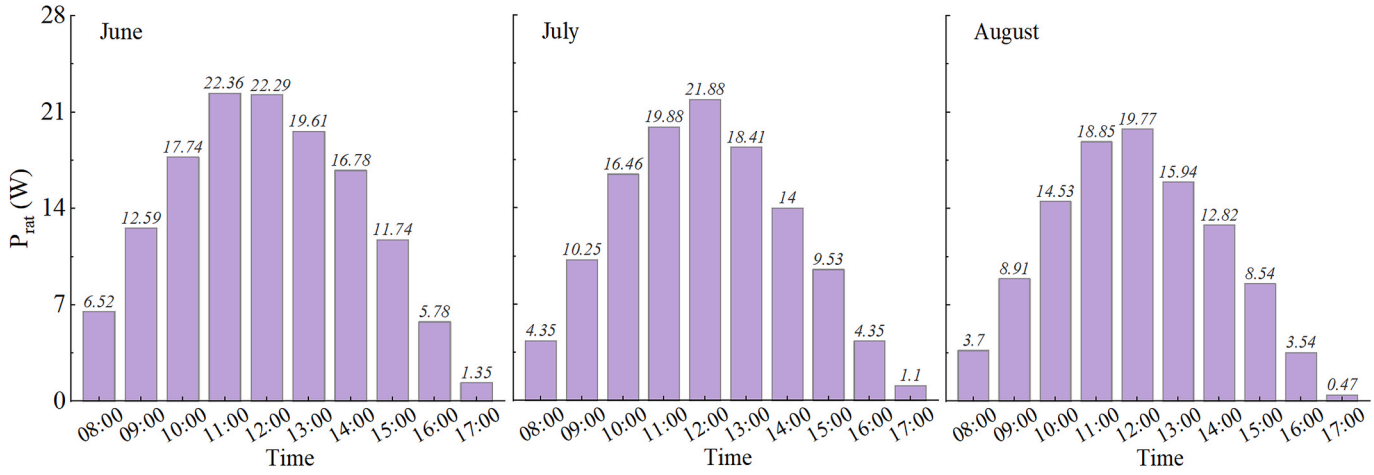


Fig. 9. Net power gain for optimal water flow cooling of photovoltaic modules.

speeds in June, which collectively enhance the evaporation rate during the cooling process.

By combining the evaporation losses with the daily P_{rat}, the water consumption per unit energy gain of the cooling system can be further quantified. The results indicate that the water-use intensity is approximately 24.3 L/kWh in June, 19.6 L/kWh in July, and 15.3 L/kWh in August. Although June exhibits the largest cooling-induced power enhancement due to stronger solar irradiance and higher module temperatures, it also requires the highest water consumption per unit electricity gain because of intensified evaporation. In contrast, the water-use efficiency improves progressively from June to August as the evaporative demand decreases. Therefore, cooling strategies must consider these monthly meteorological variations to ensure efficient and rational use of harvested rainfall.

4.4. Discussions and implications

During the operation of active water-cooling systems, the way water resources are replenished directly affects their sustainability, especially in areas with water shortages or abundant but unstable rainfall [9]. Developing low-cost, renewable water resource replenishment paths is an important discussion direction for improving the sustainability of the system. The potential synergy between evaporative water losses and natural rainfall replenishment in water cooling systems must be emphasized [55].

Photovoltaic modules have inherent rainwater harvesting potential [28,30]. Proper module inclination and surface drainage design can enhance collection efficiency [56], allowing modules to serve as composite carriers that simultaneously generate electricity and harvest rainwater [28].

To explore the passive rain harvesting capacity of photovoltaic modules under temperate monsoon conditions, this study further

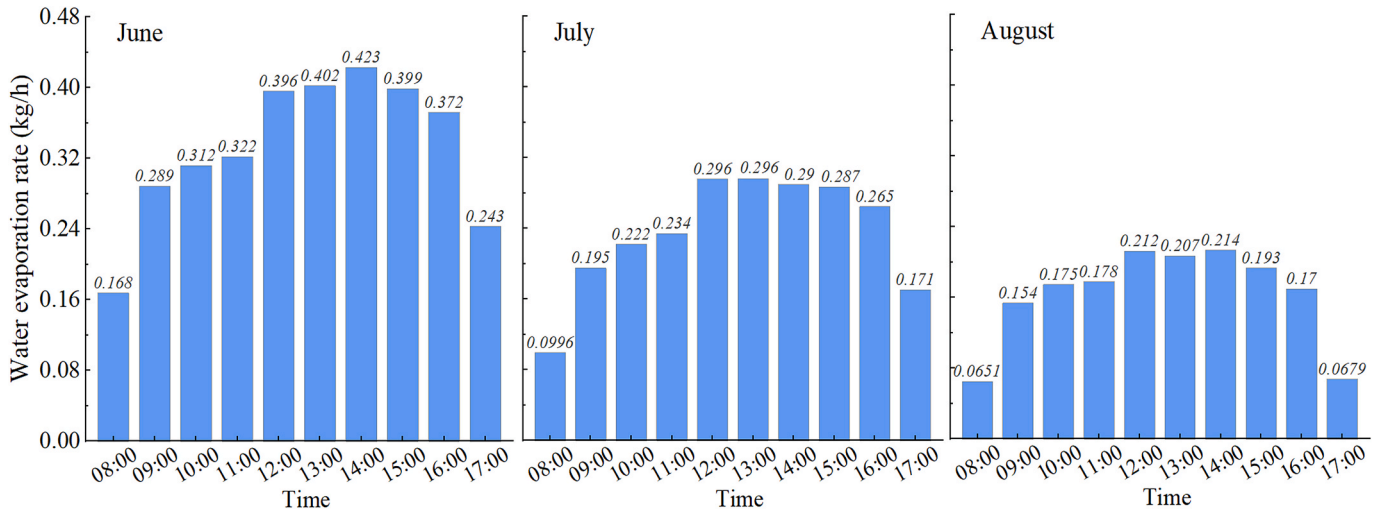


Fig. 10. Evaporation loss for optimal water flow cooling of photovoltaic modules.

conducted field tests on rooftop modules under different wind directions and inclination angles during the rainy season. As illustrated in Fig. 11, local meteorological data from the 2023–2024 rainy season indicate that light rain events accounted for 56% of rainfall occurrences, while heavy rain and more intense events comprised only 8%; however, their high precipitation intensity contributed significantly to the total rainfall volume. During precipitation events, the prevailing wind directions were predominantly from the south, southeast, and east, with these directions accounting for 66% of wind occurrence and 79% of total rainfall. Wind speeds were generally below 3 m/s, favoring stable and efficient rainwater harvesting.

Experimental results in Fig. 12 revealed that under calm conditions, the rainwater harvesting efficiency increased with module tilt angle, generally exceeding 65%. A higher tilt facilitated gravitational runoff and reduced surface water retention [56]. Under windy conditions, wind direction significantly affected droplet trajectory and thus the rainwater harvesting pathway and efficiency [57]. For windward conditions (e.g., southerly and southeasterly winds), increasing the tilt angle enhanced raindrop transport and guidance, yielding a harvesting efficiency of over 85% at a 45° tilt. In contrast, under leeward conditions (e.g., northerly or northeasterly winds), efficiency decreased with greater tilt, dropping below 70%. East and west winds showed a more neutral effect. These tests revealed the inherent rainwater harvesting capability of photovoltaic modules under natural conditions. Modules with a 35° tilt angle, particularly under southerly or southeasterly winds, not only exhibited high harvesting efficiency but also demonstrated favorable geometric suitability for solar energy capture, indicating a synergistic spatial configuration.

To further evaluate the practical relevance of rainwater harvesting for supporting photovoltaic cooling systems, the cumulative rainwater volume collected during the summer of 2024 was analyzed for different module tilt angles, as shown in Fig. 13. The results indicate that the maximum harvested rainwater volume (560.6 L) was obtained at a tilt angle of 25°, suggesting that moderate inclinations provide an effective balance between rainfall interception and gravitational runoff. For the 35° tilt angle adopted in the cooling experiments, the cumulative

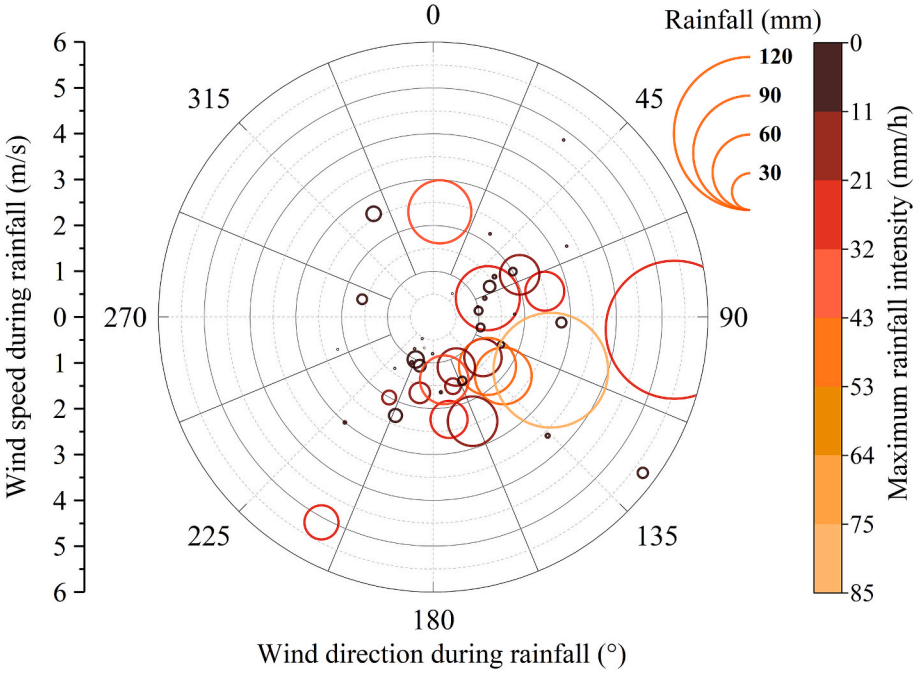


Fig. 11. Rainfall events infographic during rainy season.

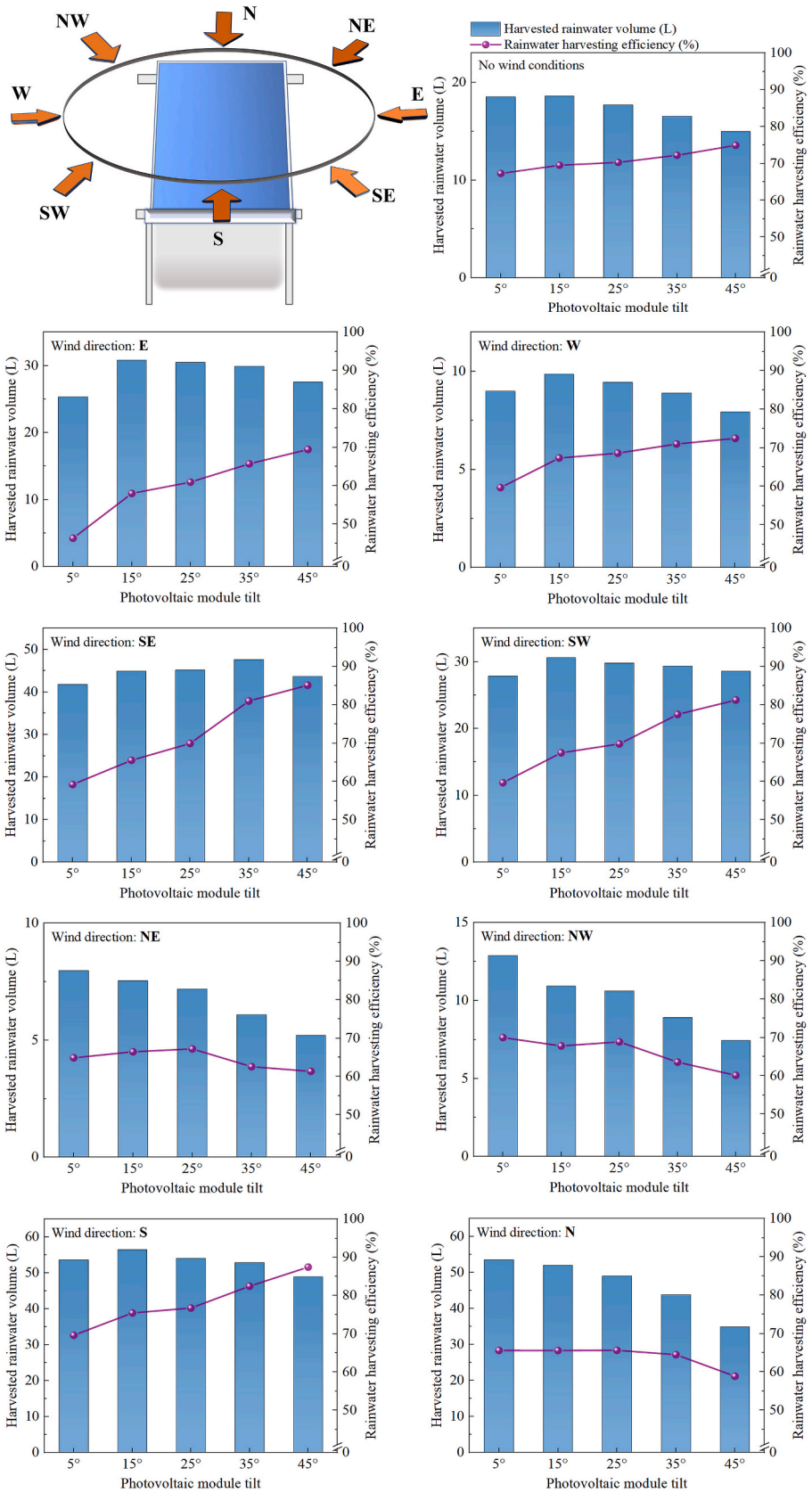


Fig. 12. Rainwater harvesting characteristics of photovoltaic modules.

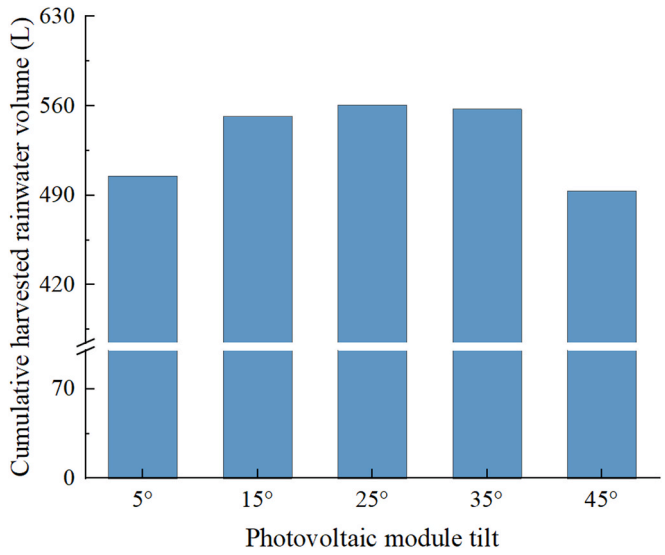


Fig. 13. Cumulative rainfall harvested by the photovoltaic module at different tilts.

harvested rainwater reached 557.5 L, which is close to the maximum value.

A comparison with the model-calculated cooling water consumption provides further insight into the potential linkage between rainwater harvesting and active cooling systems. The cumulative evaporation water consumption of the actively cooled module during the same summer period was approximately 180 L, indicating that the harvested rainwater volume is roughly two to three times the seasonal cooling water consumption. This result suggests that rooftop photovoltaic modules can function as a distributed supplementary water source for active cooling systems. However, rainfall supply and cooling demand are temporally decoupled, as rainfall occurs intermittently during precipitation events whereas cooling consumption is mainly concentrated during periods of high solar irradiance. Therefore, practical implementation would rely on appropriate rainwater storage to buffer this temporal mismatch.

The above discussion indicate that photovoltaic modules can simultaneously serve as energy conversion surfaces and rainwater collection interfaces, providing a feasible pathway for supplementing cooling water demand using naturally available precipitation. The present analysis focuses on evaluating the feasibility of linking rainwater harvesting with cooling water demand, without explicitly coupling the two processes in the current study. Future research could further investigate the dynamic matching mechanisms between rainfall availability and cooling demand, including the developing detailed water allocation or scheduling strategies to enable delayed release and multi-day utilization of harvested rainwater under varying climatic conditions.

From a broader system perspective, the practical deployment of photovoltaic water-cooling systems requires evaluation beyond the module scale. Although the present study considers the auxiliary energy consumption of the circulation pump through the net power gain indicator, a comprehensive assessment of the cumulative energy flows at the system level has not yet been conducted. Such an overall energy balance analysis would be valuable for quantifying the net energy benefit of the cooling strategy under different environmental and operational conditions.

In addition, long-term operational performance and techno-economic feasibility are also important factors affecting the practical implementation of actively cooled photovoltaic systems. Although these aspects were beyond the primary scope of the current study, the experimentally validated thermal–electrical coupled model developed in this work provides a reliable basis for future system-level investigations. By integrating long-term meteorological datasets, lifecycle performance simulations, and economic indicators such as operational costs and water resource utilization efficiency, future research could further evaluate the long-term performance stability and economic viability of the proposed cooling strategy.

5. Conclusion

This study conducted dynamic thermal–electrical modeling of a rooftop photovoltaic module integrated with an active water-cooling system, alongside experimental investigation of rainwater harvesting from the modules. The developed model elucidates the influence of environmental conditions on the water-cooling performance of photovoltaic modules and quantifies in power enhancement and evaporative water loss under optimal water flow rates. The rainwater harvesting experiments systematically evaluated the effects of module tilt and environmental factors on collection efficiency. By combining modeling and experimental analysis, this study innovatively examines both the cooling performance and the rainwater resource potential of rooftop photovoltaic systems. The principal findings are summarized as follows:

- Model validation indicates that the developed dynamic thermal–electrical coupled model has high prediction accuracy and robustness, with NRMSE values all below 5%. It can effectively capture the thermal and electrical properties of photovoltaic modules under cooling and non-cooling conditions.
- The ΔP_{rat} of active water cooling rises then falls with water flow rate, revealing an OWFR from the trade-off between cooling efficiency and pump power. Under controlled conditions, ΔP_{rat_max} relatively increases ~ 52% as irradiance increases from 800 to 1100 W/m^2 and relatively decreases 14.9–19.5% as wind speed increases from 1 to 3 m/s, indicating peak effectiveness under high thermal load and low wind.
- Seasonal analysis shows that water cooling consistently enhances photovoltaic performance under clear-sky summer conditions, with peak hourly gains up to 7.0% of rated capacity and daily gains over 100 Wh in each month. Corresponding evaporative losses decrease from 3.33 L/day in June to 1.64 L/day in August, highlighting the strong sensitivity of cooling power gain and water loss to seasonal meteorological conditions.
- Photovoltaic modules exhibit strong passive rainwater harvesting potential, typically exceeding 65% efficiency under calm conditions and surpassing 85% when winds favor runoff. Modules tilted 35°, especially under southerly or southeasterly winds, combine high rainwater harvesting efficiency with effective solar energy capture.

This study supports the possibility of using harvested rainwater to supplement active photovoltaic cooling, with seasonal rainwater harvested volumes exceeding cooling water consumption by two to three times. While rainfall and cooling needs are temporally decoupled, appropriate storage and scheduling can enable effective utilization. Future research should explore dynamic integration of rainwater harvesting, storage, and cooling, along with system-level energy and economic assessments, to optimize water use, enhance autonomy, and support sustainable deployment across diverse rooftops and climates.

CRediT authorship contribution statement

Runze Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Chao Ma:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Conceptualization. **Zijian Xu:** Validation, Investigation, Data curation. **Wenchao Qi:** Visualization, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

References

[1] Hamed MH, Hassan H, Ookawara S, et al. PV thermal regulation and water harvesting hybrid system using evaporative cooling of silica gel layer. Process Saf Environ Prot 2024;181:243–65. https://doi.org/10.1016/j.psep.2023.11.009.
[2] Rahmani A, Hosseinifard F, Salimi M, et al. A hybrid machine learning approach for predicting the performance of perovskite solar cells under varying temperatures. Energy Convers Manage: X 2025;101217. https://doi.org/10.1016/j.ecmx.2025.101217.
[3] Lopez-Pascual D, Valiente-Blanco I, Fernandez-Munoz M, et al. Theoretical modelling and optimization of a geothermal cooling system for solar photovoltaics. Renew Energy 2023;206:357–66. https://doi.org/10.1016/j.renene.2023.01.098.
[4] Bevilacqua P, Morabito A, Bruno R, et al. Seasonal performances of photovoltaic cooling systems in different weather conditions. J Clean Prod 2020;272:122459. https://doi.org/10.1016/j.jclepro.2020.122459.
[5] Dwivedi P, Sudhakar K, Soni A, et al. Advanced cooling techniques of PV modules: a state of art. Case Stud Therm Eng 2020;21:100674. https://doi.org/10.1016/j.csite.2020.100674.
[6] Skoplaki E, Palyvos JA. On the temperature dependence of photovoltaic module electrical performance: a review of efficiency/power correlations. Sol Energy 2009;83(5):614–24. https://doi.org/10.1016/j.solener.2008.10.008.
[7] Muneeshwaran M, Sajjad U, Ahmed T, et al. Performance improvement of photovoltaic modules via temperature homogeneity improvement. Energy 2020;203:117816. https://doi.org/10.1016/j.energy.2020.117816.
[8] Vats K, Tiwari GN. Energy and exergy analysis of a building integrated semitransparent photovoltaic thermal (BISPVT) system. Appl Energy 2012;96:409–16. https://doi.org/10.1016/j.apenergy.2012.02.079.
[9] Koohestani SS, Nižetić S, Santamouris M. Comparative review and evaluation of state-of-the-art photovoltaic cooling technologies. J Clean Prod 2023;406:136953. https://doi.org/10.1016/j.jclepro.2023.136953.
[10] Gharzi M, Arabhosseini A, Gholami Z, et al. Progressive cooling technologies of photovoltaic and concentrated photovoltaic modules: a review of fundamentals, thermal aspects, nanotechnology utilization and enhancing performance. Sol Energy 2020;211:117–46. https://doi.org/10.1016/j.solener.2020.09.048.
[11] Čabo FG, Nižetić S, Giama E, et al. Techno-economic and environmental evaluation of passive cooled photovoltaic systems in Mediterranean climate conditions. Appl Therm Eng 2020;169:114947. https://doi.org/10.1016/j.applthermaleng.2020.114947.
[12] Hernandez-Perez JG, Carrillo JG, Bassam A, et al. A new passive PV heatsink design to reduce efficiency losses: a computational and experimental evaluation. Renew Energy 2020;147:1209–20. https://doi.org/10.1016/j.renene.2019.09.088.
[13] Shrivastava A, Jose JPA, Borole YD, et al. A study on the effects of forced air-cooling enhancements on a 150 W solar photovoltaic thermal collector for green cities. Sustainable Energy Technol Assess 2022;49:101782. https://doi.org/10.1016/j.seta.2021.101782.
[14] Schiro F, Benato A, Stoppato A, et al. Improving photovoltaics efficiency by water cooling: modelling and experimental approach. Energy 2017;137:798–810. https://doi.org/10.1016/j.energy.2017.04.164.
[15] Reddy SR, Ebadian MA, Lin CX. A review of PV–T systems: thermal management and efficiency with single phase cooling. Int J Heat Mass Transf 2015;91:861–71. https://doi.org/10.1016/j.ijheatmasstransfer.2015.07.134.
[16] Nižetić S, Čoko D, Yadav A, et al. Water spray cooling technique applied on a photovoltaic panel: the performance response. Energ Conver Manage 2016;108:287–96. https://doi.org/10.1016/j.enconman.2015.10.079.
[17] Hadipour A, Zargarabadi MR, Rashidi S. An efficient pulsed-spray water cooling system for photovoltaic panels: Experimental study and cost analysis. Renew Energy 2021;164:867–75. https://doi.org/10.1016/j.renene.2020.09.021.
[18] Bevilacqua P, Bruno R, Rollo A, et al. A novel thermal model for PV panels with back surface spray cooling. Energy 2022;255:124401. https://doi.org/10.1016/j.energy.2022.124401.
[19] Da Silva VO, Martinez-Bolanos JR, Heideier RB, et al. Theoretical and experimental research to development of water-film cooling system for commercial photovoltaic modules. IET Renew Power Gener 2021;15(1):206–24. https://doi.org/10.1049/rpg2.12017.
[20] Azimi A, Basiri N, Eslami M. A novel two-step optimization approach for film water cooling of a photovoltaic module in real ambient conditions. Appl Therm Eng 2024;236:121898. https://doi.org/10.1016/j.applthermaleng.2023.121898.
[21] Shahverdian MH, Sohani A, Sayyaadi H. Water-energy nexus performance investigation of water flow cooling as a clean way to enhance the productivity of solar photovoltaic modules. J Clean Prod 2021;312:127641. https://doi.org/10.1016/j.jclepro.2021.127641.
[22] Shahverdian MH, Sohani A, Sayyaadi H, et al. A dynamic multi-objective optimization procedure for water cooling of a photovoltaic module. Sustainable Energy Technol Assess 2021;45:101111. https://doi.org/10.1016/j.seta.2021.101111.
[23] Yang LH, Liang JD, Hsu CY, et al. Enhanced efficiency of photovoltaic panels by integrating a spray cooling system with shallow geothermal energy heat exchanger. Renew Energy 2019;134:970–81. https://doi.org/10.1016/j.renene.2018.11.089.
[24] Zhang Q, He S, Song T, et al. Modeling of a PV system by a back-mounted spray cooling section for performance improvement. Appl Energy 2023;332:120532. https://doi.org/10.1016/j.apenergy.2022.120532.
[25] Ismael ORI, Selimli S. Energy and exergy analysis of PV modules: Sustainable improvements through fin structure and spray cooling. Sol Energy Mater Sol Cells 2025;283:113458. https://doi.org/10.1016/j.solmat.2025.113458.
[26] Dida M, Boughali S, Bechki D, et al. Experimental investigation of a passive cooling system for photovoltaic modules efficiency improvement in hot and arid regions. Energ Conver Manage 2021;243:114328. https://doi.org/10.1016/j.enconman.2021.114328.
[27] Yao W, Kong X, Xu A, et al. New models for the influence of rainwater on the performance of photovoltaic modules under different rainfall conditions. Renew Sustain Energy Rev 2023;173:113119. https://doi.org/10.1016/j.rser.2022.113119.
[28] Myyas RN, Al-Dabbasa M, Tostado-Véliz M, et al. A novel solar panel cleaning mechanism to improve performance and harvesting rainwater. Sol Energy 2022;237:19–28. https://doi.org/10.1016/j.solener.2022.03.068.
[29] García-Chica A, Rodriguez-Perez AM, Caparros-Mancera JJ, et al. Integrated photovoltaic system for rainwater collection and sustainable irrigation. Irrig Sci 2025;43(6):1385–95. https://doi.org/10.1007/s00271-025-01028-7.
[30] Chekired F, Richa A, Touil S, et al. Energy yield evaluation of a rainwater harvesting system using a novel agrophotovoltaics design. Desalin Water Treat 2022;255:27–33. https://doi.org/10.5004/dwt.2022.28318.
[31] Arshad R, de la Loma GB, Monente GS. Rainwater harvesting potential from photovoltaic energy systems in the Sahel. Water-Energy Nexus 2025;8:115–31. https://doi.org/10.1016/j.wen.2025.04.002.
[32] Alazzam I, Shatanawi K, Al-Weshah R. Rainwater and fog harvesting from solar panels. Global Journal of Environmental Science and Management, 2024, 10(3):955-968. Doi: 10.22035/gjesm.2024.03.
[33] Armstrong S, Hurley WG. A thermal model for photovoltaic panels under varying atmospheric conditions. Appl Therm Eng 2010;30(11–12):1488–95. https://doi.org/10.1016/j.applthermaleng.2010.03.012.
[34] Yaman K, Arslan G. A detailed mathematical model and experimental validation for coupled thermal and electrical performance of a photovoltaic (PV) module. Appl Therm Eng 2021;195:117224. https://doi.org/10.1016/j.applthermaleng.2021.117224.
[35] Rahaman MA, Chambers TL, Fekih A, et al. Floating photovoltaic module temperature estimation: modeling and comparison. Renew Energy 2023;208:162–80. https://doi.org/10.1016/j.renene.2023.03.076.
[36] Swinbank WC. Long-wave radiation from clear skies. Q J R Meteorolog Soc 1963;89(381):339–48. https://doi.org/10.1002/qj.49708938105.
[37] Incropera FP, DeWitt DP. Fundamentals of heat and mass transfer[M]. New York: Wiley; 1996. p. 346–58.
[38] Zhao Z N. Heat transfer 3rd edition[M]. Beijing: Higher Education Press, 2019, pp. 218-223, 257-275.
[39] Kreith F, Bohn M S. Principles of heat transfer (Fourth edition)[M]. New York: Harper & Row, 1986, pp. 209-219.
[40] Kjeldstad T, Lindholm D, Marstein E, et al. Cooling of floating photovoltaics and the importance of water temperature. Sol Energy 2021;218:544–51. https://doi.org/10.1016/j.solener.2021.03.022.
[41] Poós T, Varju E. Mass transfer coefficient for water evaporation by theoretical and empirical correlations. Int J Heat Mass Transf 2020;153:119500. https://doi.org/10.1016/j.ijheatmasstransfer.2020.119500.
[42] Aït Saada S, Nebbali R, Kecili I, et al. Operating temperature and electrical efficiency of a photovoltaic panel cooled by laminar water film flowing on its front face: Experimental and numerical approach. Mater Today Sustainability 2025;30:101093. https://doi.org/10.1016/j.mtsust.2025.101093.
[43] Fulford GD. The flow of liquids in thin films[M]//Advances in chemical engineering. Academic Press 1964;5:151–236. https://doi.org/10.1016/S0065-2377(08)60008-3.
[44] Li H, Zhao J, Li M, et al. Performance analysis of passive cooling for photovoltaic modules and estimation of energy-saving potential. Sol Energy 2019;181:70–82. https://doi.org/10.1016/j.solener.2019.01.014.
[45] Chakrasali RL, Sheelavant VR, Nagaraja HN. Network approach to modeling and simulation of solar photovoltaic cell. Renew Sustain Energy Rev 2013;21:84–8. https://doi.org/10.1016/j.rser.2013.01.001.
[46] Sohani A, Cornaro C, Shahverdian MH, et al. Thermography and machine learning combination for comprehensive analysis of transient response of a photovoltaic

module to water cooling. Renew Energy 2023;210:451–61. https://doi.org/10.1016/j.renene.2023.04.073.
[47] Larock BE, Jeppson RW, Watters GZ. Hydraulics of pipeline systems[M]. CRC Press; 1999.
[48] Yu X, Song Z, Ding X, et al. Study on deep learning-based rural residential rooftop space extraction and photovoltaic power generation potential—A case study of Baodi district. Tianjin. Building Science 2025;41:95–103. https://doi.org/10.13614/j.cnki.11-1962/tu.2025.10.10.
[49] Liu Y, Yao L, Jiang H, et al. Spatial estimation of the optimum PV tilt angles in China by incorporating ground with satellite data. Renew Energy 2022;189:1249–58. https://doi.org/10.1016/j.renene.2022.03.072.
[50] Zhao W, Zhang X, Yang S, et al. Assessment of long-term photovoltaic (PV) power potential in China based on high-quality solar radiation and optimal tilt angles of PV panels. Remote Sens (Basel) 2025;17(18):3235. https://doi.org/10.3390/rs17183235.
[51] Elminshawy NAS, Elminshawy A, Osama A. An innovative cooling technique for floating photovoltaic module: Adoption of partially submerged angle fins. Energy Convers Manage: X 2023;20:100408. https://doi.org/10.1016/j.ecmx.2023.100408.
[52] Ahn H, Rim D, Pavlak GS, et al. Uncertainty analysis of energy and economic performances of hybrid solar photovoltaic and combined cooling, heating, and power (CCHP+ PV) systems using a Monte-Carlo method. Appl Energy 2019;255:113753. https://doi.org/10.1016/j.apenergy.2019.113753.
[53] Lu S, Liang R, Zhang J, et al. Performance improvement of solar photovoltaic/thermal heat pump system in winter by employing vapor injection cycle. Appl Therm Eng 2019;155:135–46. https://doi.org/10.1016/j.applthermaleng.2019.03.038.
[54] Akhsassi M, Erraissi N, Raoufi M, et al. Experimental investigation and thermo-electrical performance modeling of two PV plants in arid climates. Energ Conver Manage 2024;321:119025. https://doi.org/10.1016/j.enconman.2024.119025.
[55] Englart S. Analysis of rainwater use in membrane-based semi-direct evaporative cooling of air. J Build Eng 2024;90:109409. https://doi.org/10.1016/j.jobe.2024.109409.
[56] Kord SMZ, Tuure J, Räsänen M, et al. Geometry and surface manipulation impact on passive dew and rain collection. Agric For Meteorol 2024;356:110180.
[57] De Souza TD, Ghisi E. Harvesting rainwater from scaffolding platforms and walls to reduce potable water consumption at buildings construction sites. J Clean Prod 2020;258:120909. https://doi.org/10.1016/j.jclepro.2020.120909.